Back to Configure the Stations

Back to Program the Method that Places Parts into Stock

Back to Program the Method that Removes Parts from Stock

Back to Program the Method that Manages the Inventory Table

Back to Display a Value During the Simulation with a Variable

Back to Place Parts into Stock and Remove Parts from It

Using the Store as Supermarket

Use the Store as Supermarket

The *Store* can also work as supermarket and thus represent a pulling material flow.

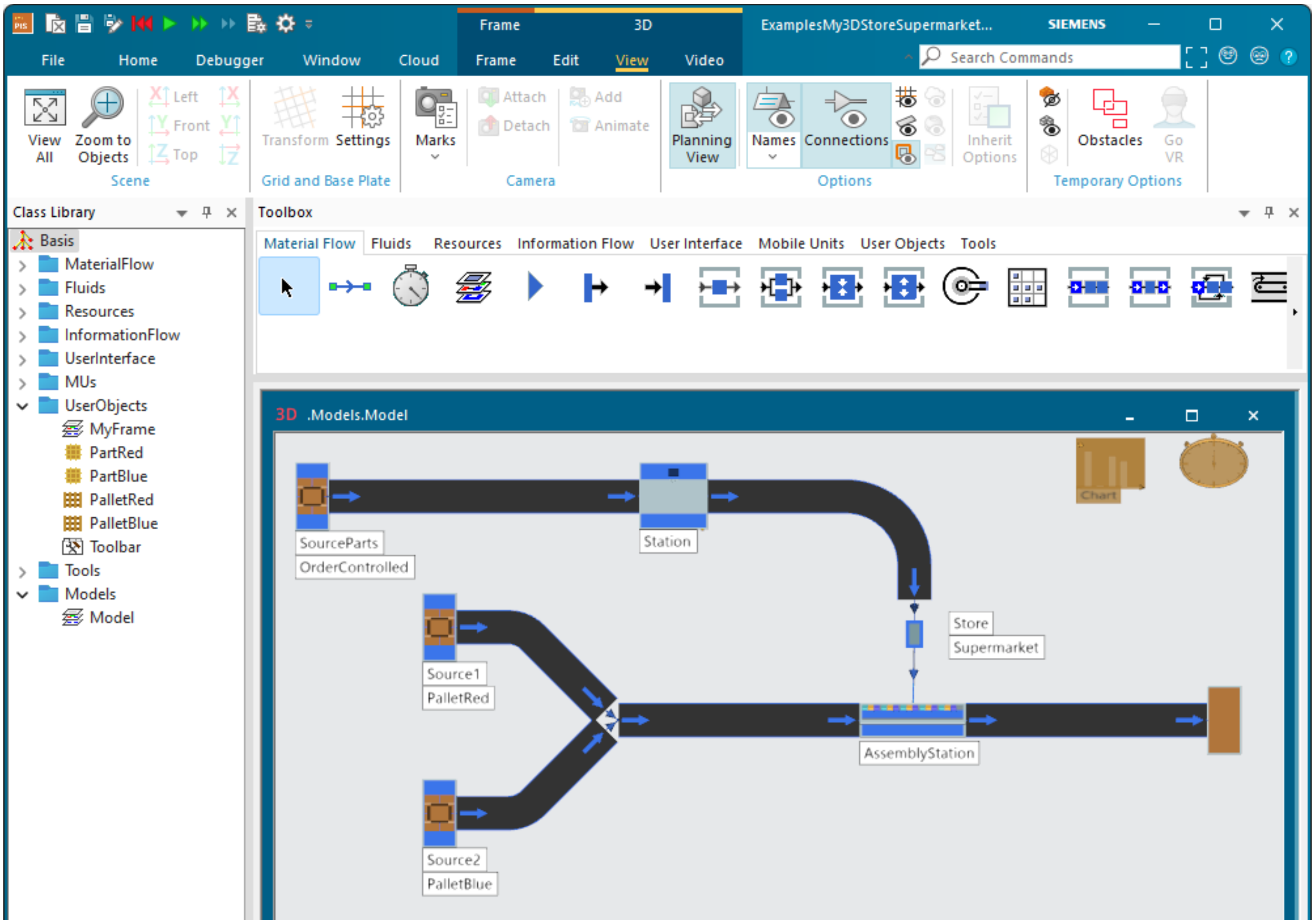
Here we show how the **Store** orders two different part types from a *Source*. Only when the stock of these parts falls below a certain value, the *Source* starts producing these parts. To facilitate this, you have to activate **Supermarket** in the *Store* and **MU Selection > Order Controlled** in the *Source*.

The *Store* then sends these parts on to an *AssemblyStation*, which loads them onto the respective pallets and moves them on.

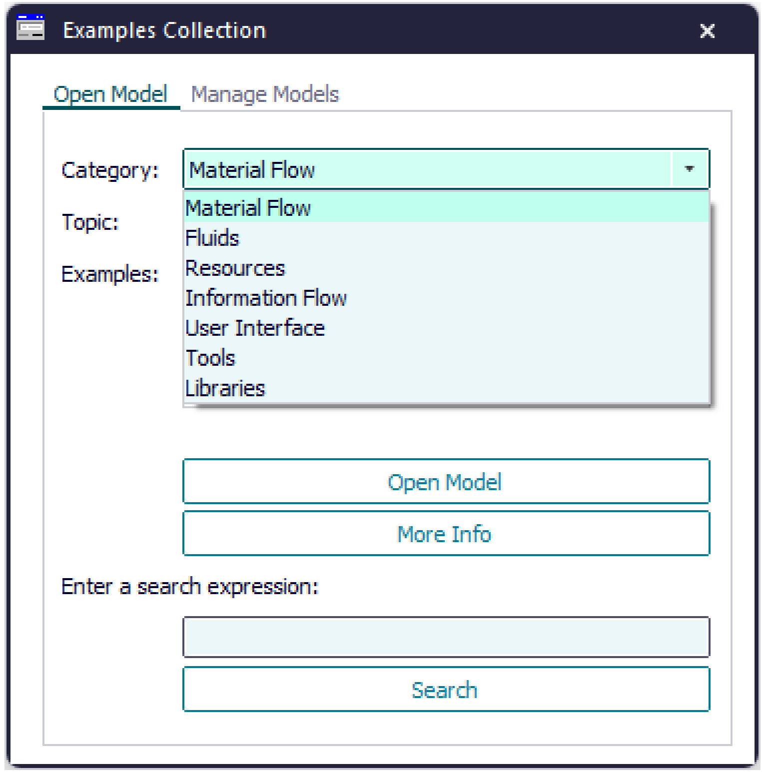
We will demonstrate how to:

- **Model the Pulling Material Flow**
- **Configure the Store as Supermarket and the Source as Order-Controlled**
- **Show the Amount of Waiting Parts and the Amount of Orders**

The finished simulation model looks like this:



Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



Compare Place Parts into Stock and Remove Parts from It

Compare Stack Parts in the Store

Back to Model the Flow of Materials, Basics

Go to Show the Amount of Waiting Parts and the Amount of Orders

Video on YouTube

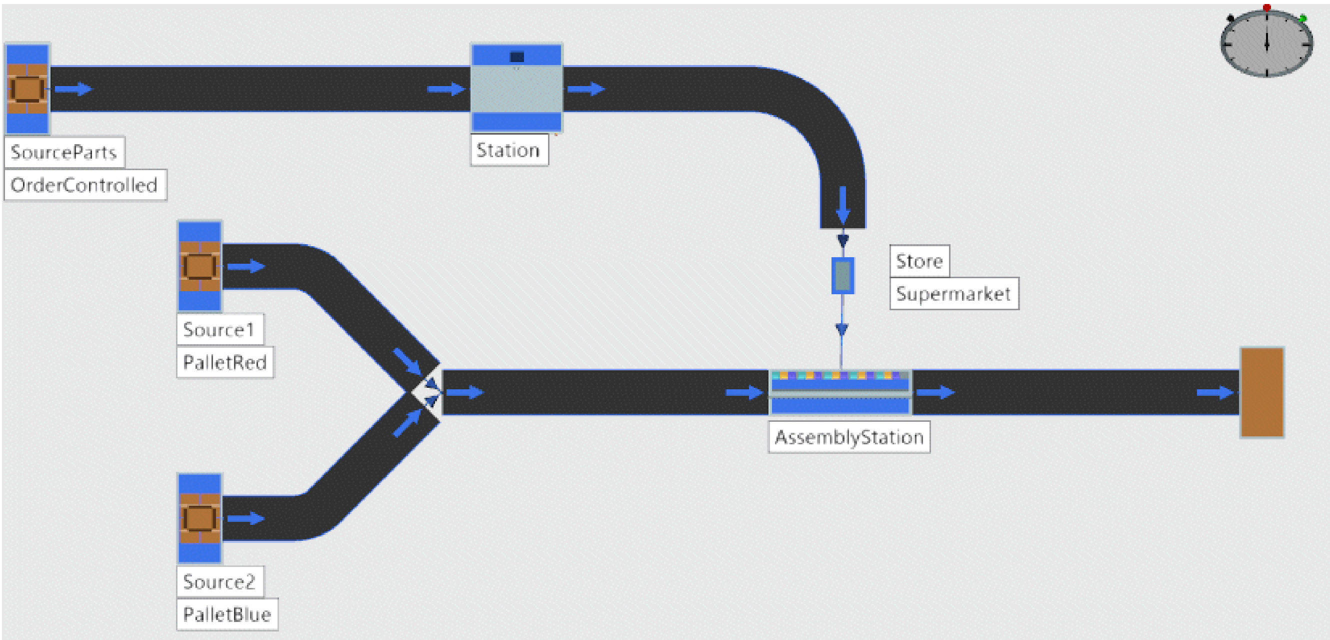
https://youtu.be/ISYfZMdQp3w?si=oRZapujKzN7zp5-_&t=381

Model the Pulling Material Flow

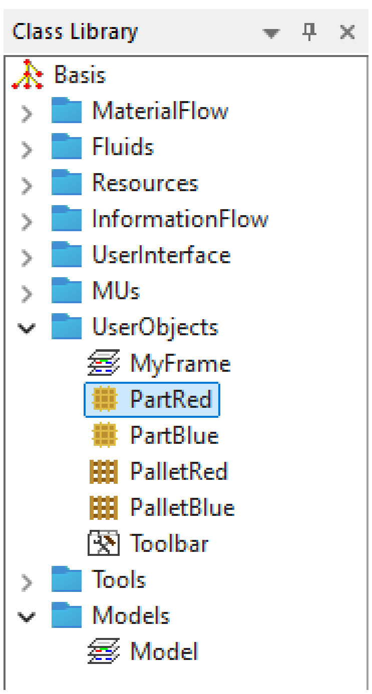
When the *Store* works as supermarket, we model the pulling material flow.

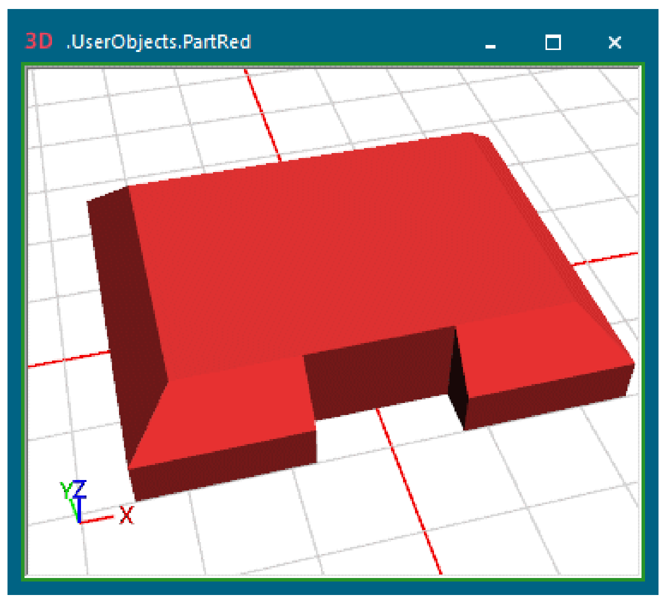
To model the pulling material flow:

- Insert the material flow objects so that they match the figure below.

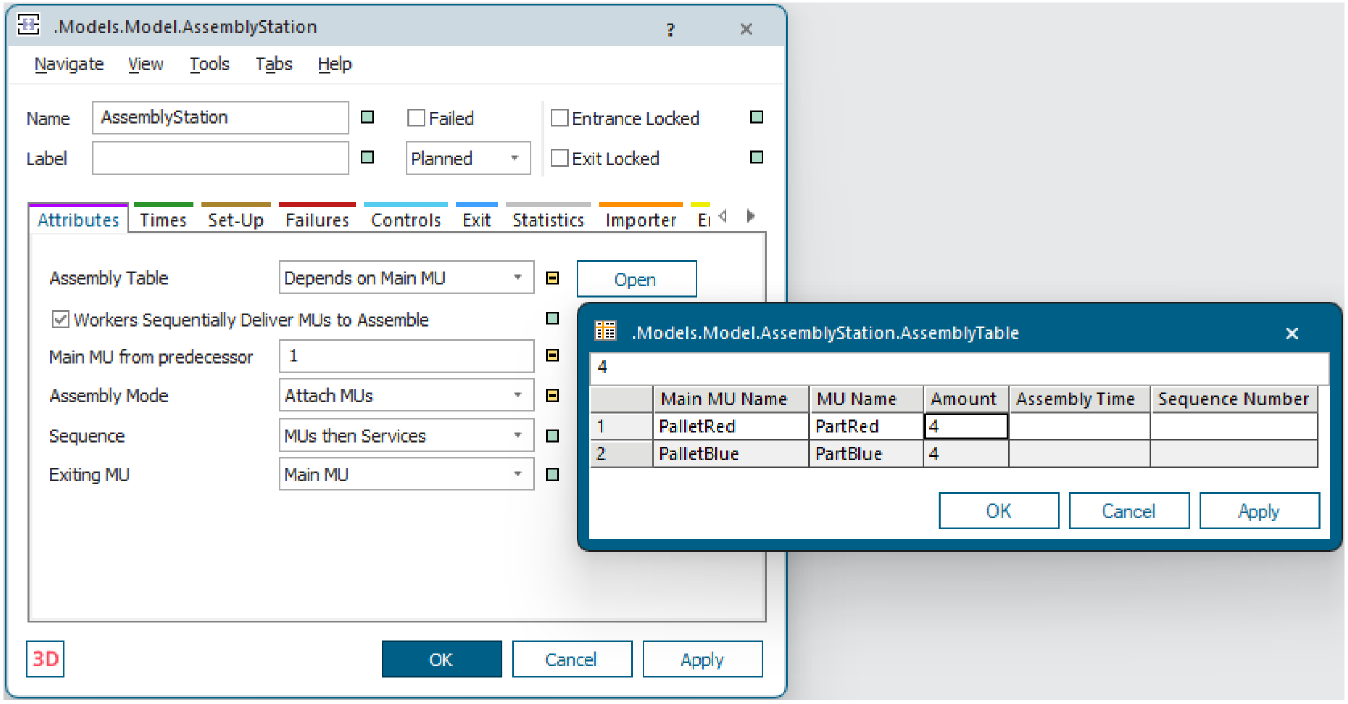


- Duplicate the *Part* and the *Container* twice each. We renamed our *Parts* to `PartRed` and `PartBlue` and our *Containers* to `PalletRed` and `PalletBlue` and colored them accordingly. We did not change any of the other settings.



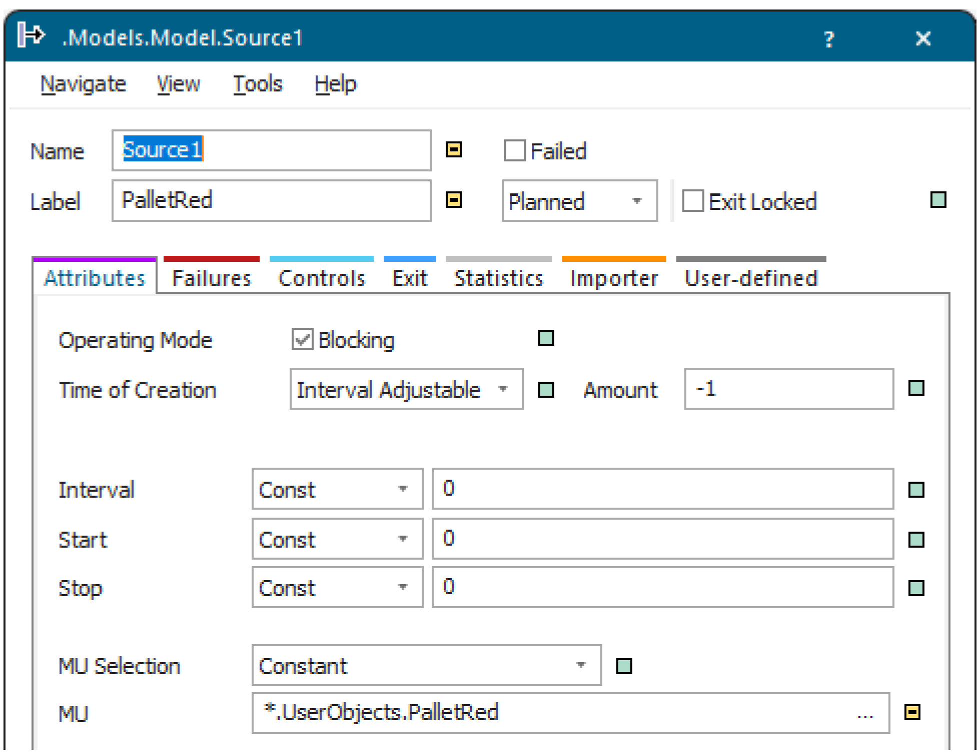


- Shorten the **Processing Time** of the *Station* from **10** to 2 (0:02) seconds.
- Configure the *AssemblyStation* as shown in the figure below. You only have to enter the names of the parts, not their path.



Shorten the **Processing Time** of the *AssemblyStation* from **10** to 5 (0:05) seconds.

- Enter the respective *pallets* into *Source1* and *Source2*.



Back to **Use the Store as Supermarket**

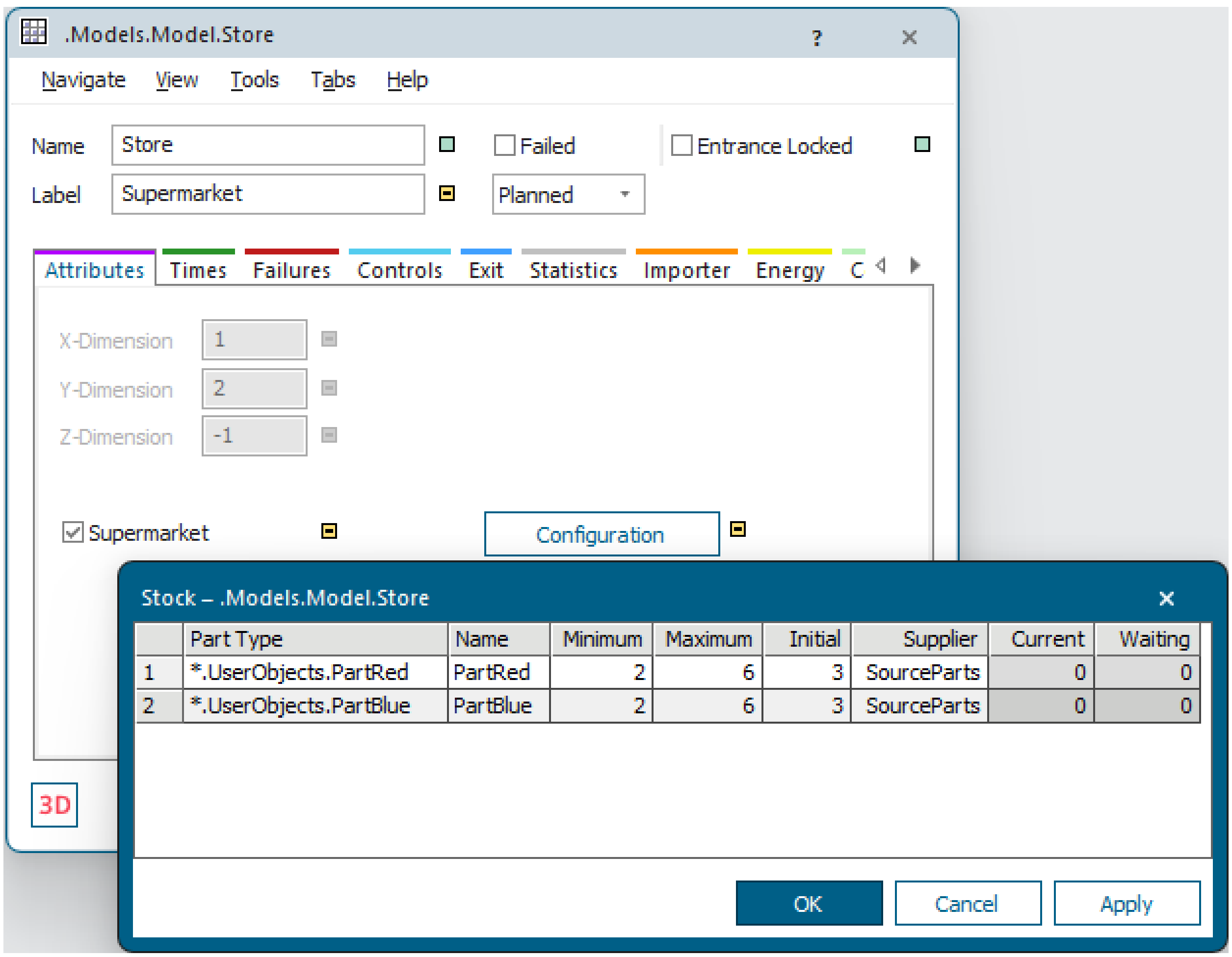
Configure the Store as Supermarket and the Source as Order-Controlled

Configure the *Store* as **Supermarket** and the *Source* as **Order-Controlled** in the simulation model.

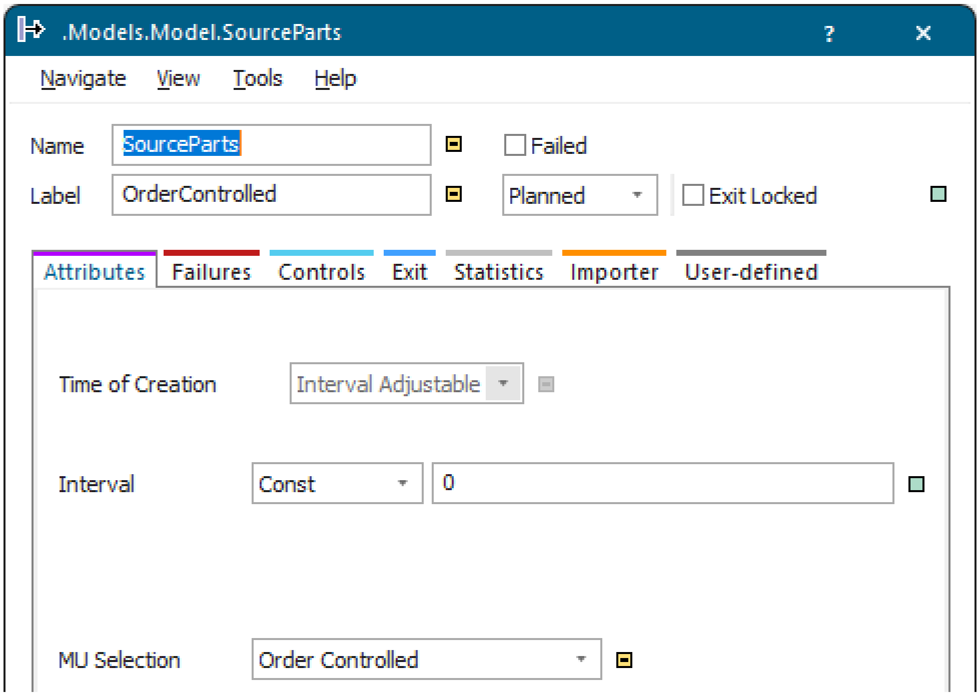
To configure the *Store* as **Supermarket** and the *Source* as **Order Controlled**:

Open the dialog of the *Store*:

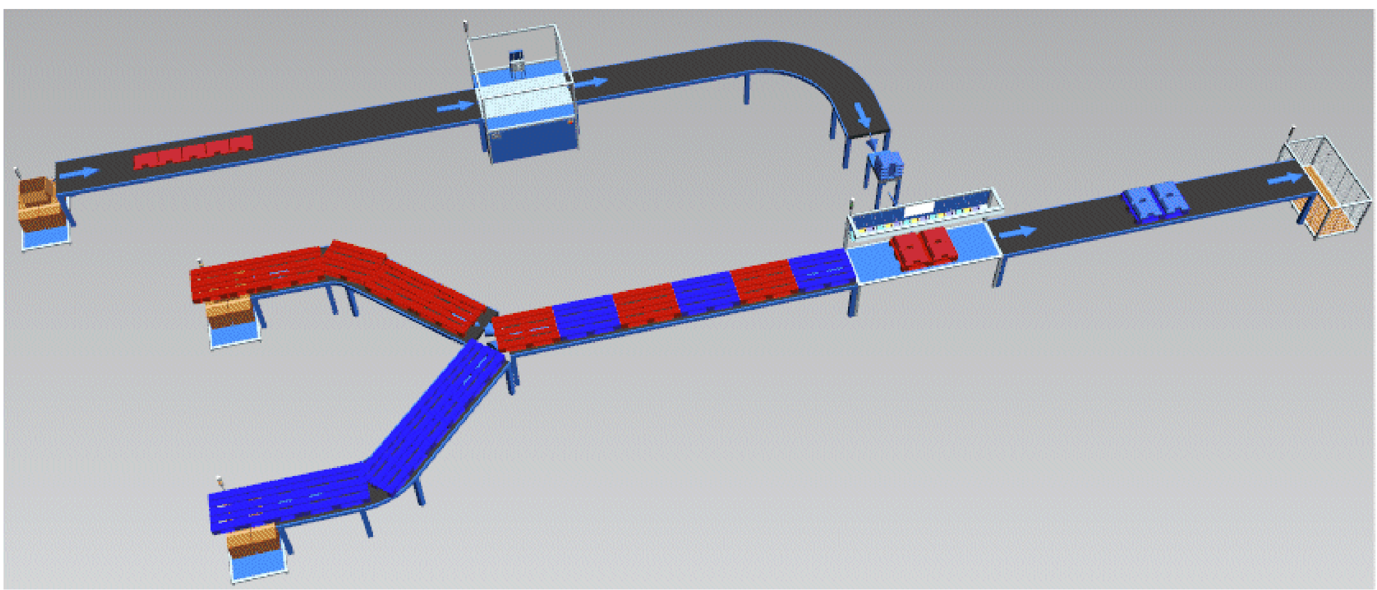
- Select the check box **Supermarket**.
- Click **Configuration** and fill in the configuration table. Our *Store* has a **Minimum** stock of 2 parts for our *PartRed* and *PartBlue*, a **Maximum** stock of 6 parts, and an **Initial** stock of 3 parts. The **Supplier** of the parts is the *Source* named *SourceParts*. During the simulation the column **Current** shows how many parts are located in the *Store*.



- Open the dialog of the *Source* and select **MU selection > Order Controlled**.



Run the simulation and watch what happens.



> **Note**
>
> If the simulation does not start, check the setting **Main part from predecessor no** of the *AssemblyStation* and change it if necessary.

Back to Use the Store as Supermarket

Show the Amount of Waiting Parts and the Amount of Orders

Show the amount of waiting parts and the amount of orders.

You can watch that the *Store* initially contains 3 parts each, which are loaded onto the pallets. Once the **Minimum** stock is reached, the *Store* reorders parts from the *Source*, which only then starts producing them. These parts are then fed into the material flow.

Stock – .Models.Model.Store

	Part Type	Name	Minimum	Maximum	Initial	Supplier	Current	Waiting
1	*.UserObjects.PartRed	PartRed	2	6	3	SourceParts	0	4
2	*.UserObjects.PartBlue	PartBlue	2	6	3	SourceParts	1	5

The command **View > Show Orders** of the *Source* shows the orders which are pending with this *Source*. The table contains the **Part Type** of the ordered *MUs*, its **Amount**, and the **Target** of the part. The **Target** is the object that orders the parts.

.Models.Model.SourceParts.Orders

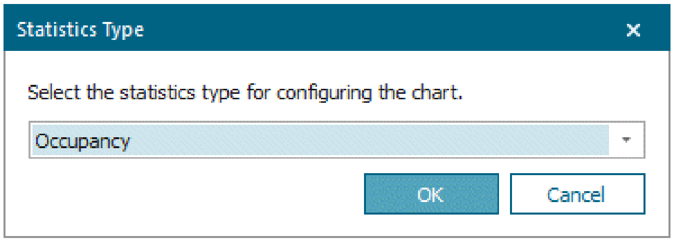
	Name	Amount	Target
1	PartRed	5	*.Models.Model.Store

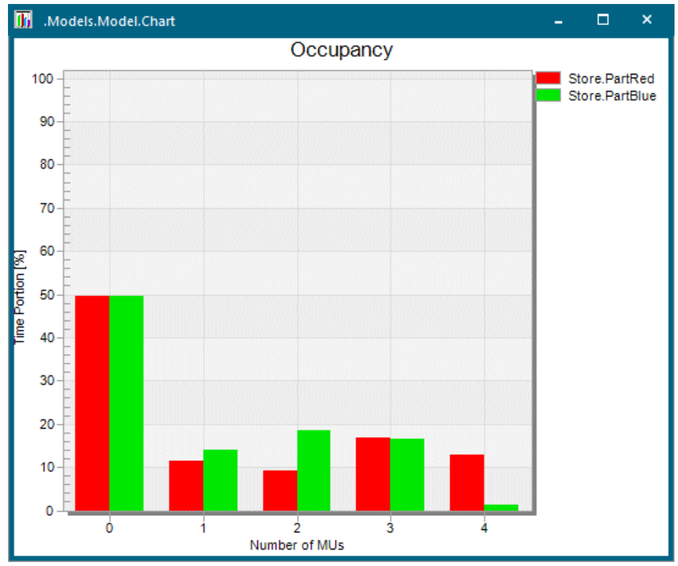
The command **View > Show Orders** of the *Store* shows the orders which are pending with this *Store*.

.Models.Model.Store.Orders

	Name	Amount	Target
1	PartRed	1	*.Models.Model.AssemblyStation

When you drag the *Store* configured as a **Supermarket** onto a *Chart*, it shows the **occupancy** for all parts that are defined in the **Configuration** table by default.





Back to **Use the Store as Supermarket**

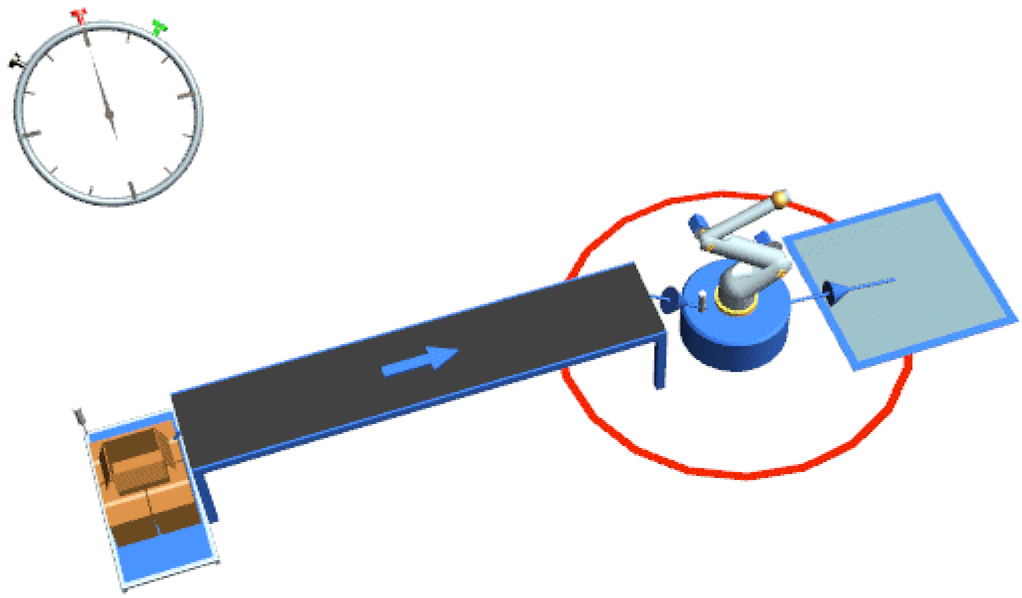
Stack Parts in the Store

When you model your plant, you can also stack parts in the *Store*.

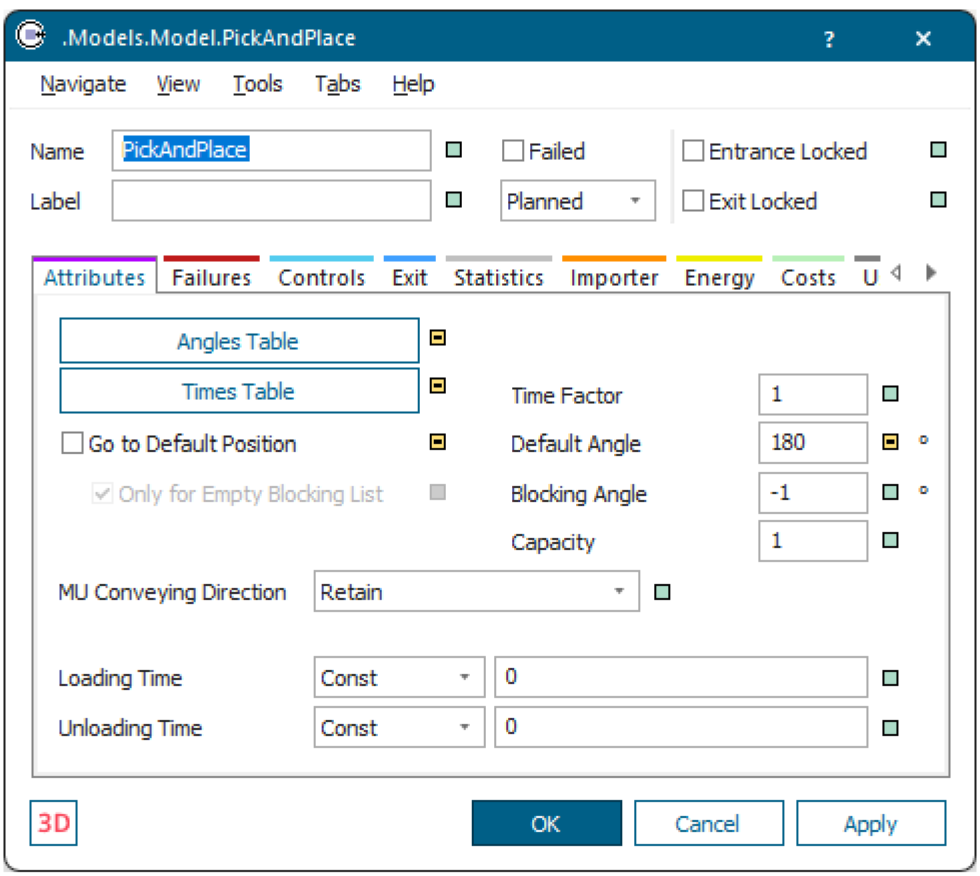
To do so, just enter the **Z-Dimension** of the **Store**.

In our example the *Source* produces parts and moves them on onto a *Conveyor*. The *Robot* then unloads the parts from the *Conveyor* and places them in the *Store*.

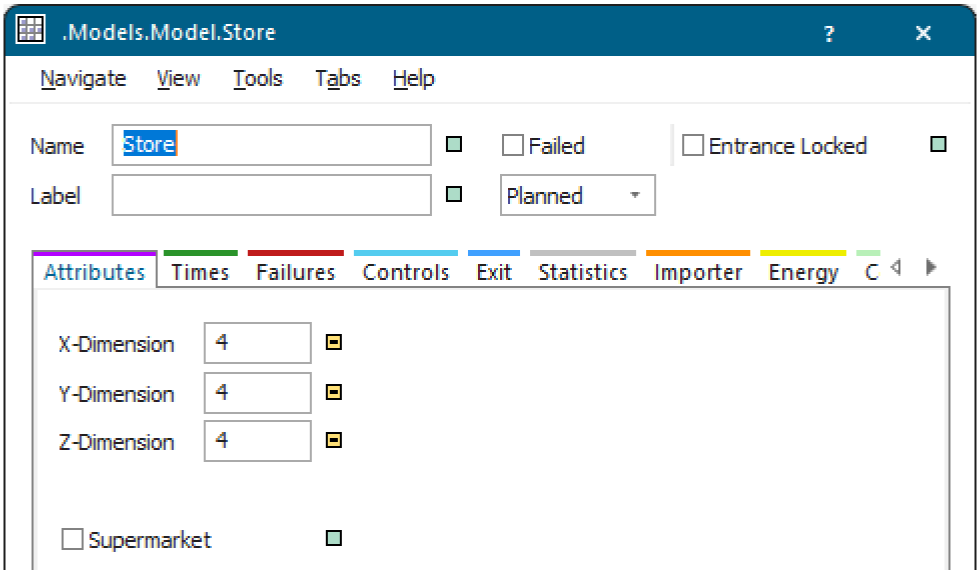
The finished simulation model looks like this:



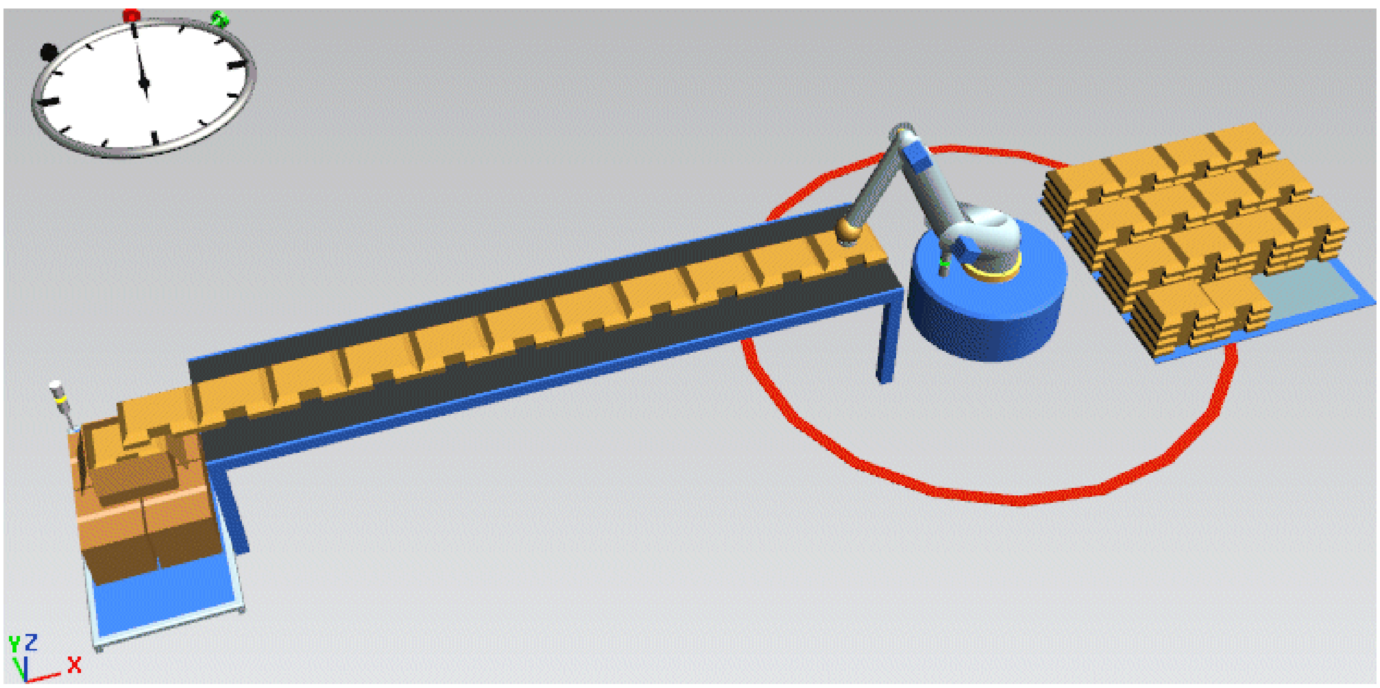
- For the *Source* and the *Conveyor* we used the default settings.
- For the *Robot* we entered a **Loading Time** and an **Unloading Time** of 1 second (`0:01`) each and a **Default Angle** of `180` degrees.



- Our *Store* can store 4 parts in each dimension.



When we run the simulation, we can watch how the *Robot* places the parts into stock and stacks them. The screenshots also shows that the parts move onto the *Conveyor* continually with the front first.



Compare Unload Stacked Parts

Compare Define the Animation Area of a Simulation Object

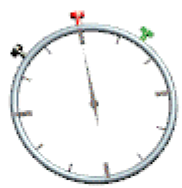
Back to Model the Flow of Materials, Basics

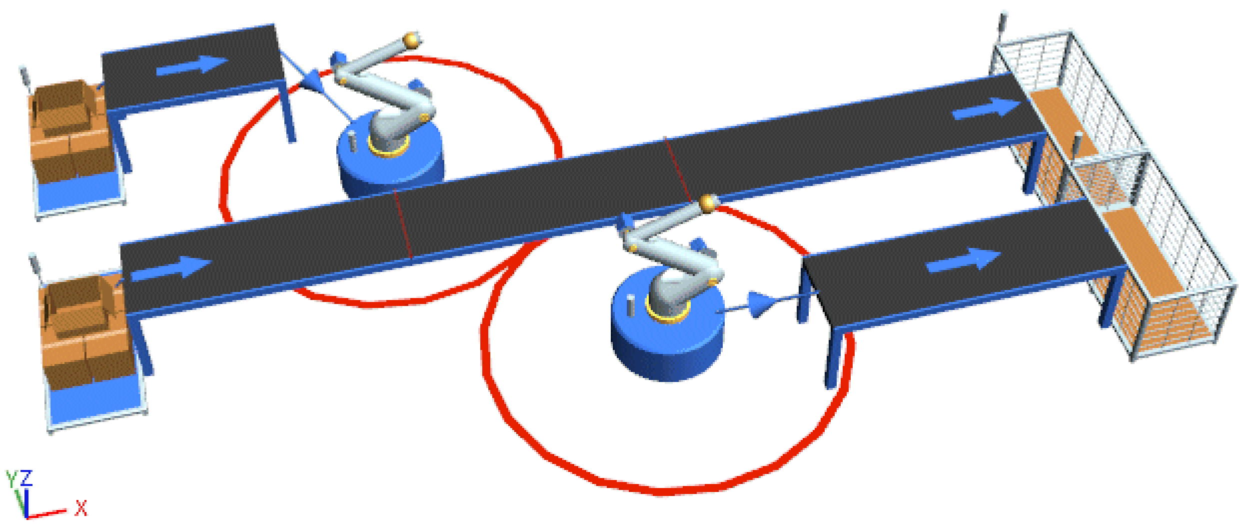
Loading a Container Layer by Layer

By default, *Plant Simulation* stacks the parts on each loading place of the *Container* to its maximum **Z-Dimension** and then starts a new layer on a new place.

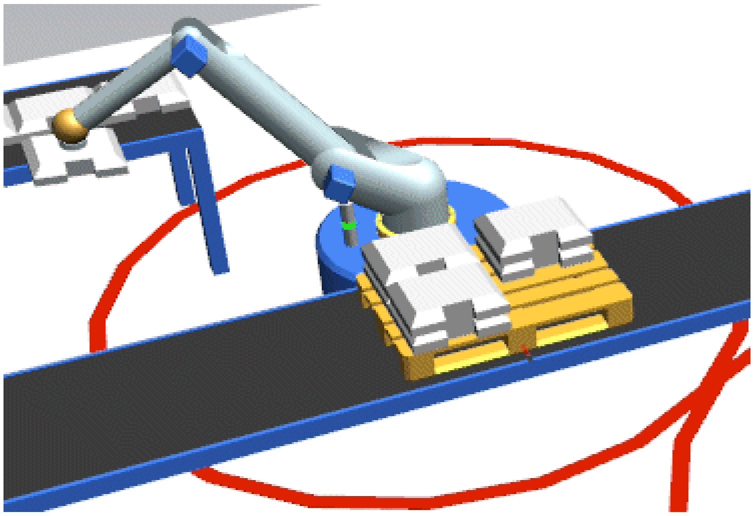
You can also load the parts layer by layer by activating **Fill Whole Layer** ☑ Fill Whole Layer .

In our sample model a *Source* produces parts and another *Source* produces pallets. The top robot loads the parts onto the pallet at the first sensor of the *Conveyor*. The bottom robot unloads the parts from the pallet and places them onto the lower *Conveyor*. The *Conveyors* move the pallets and the parts on to their respective *Drains*.

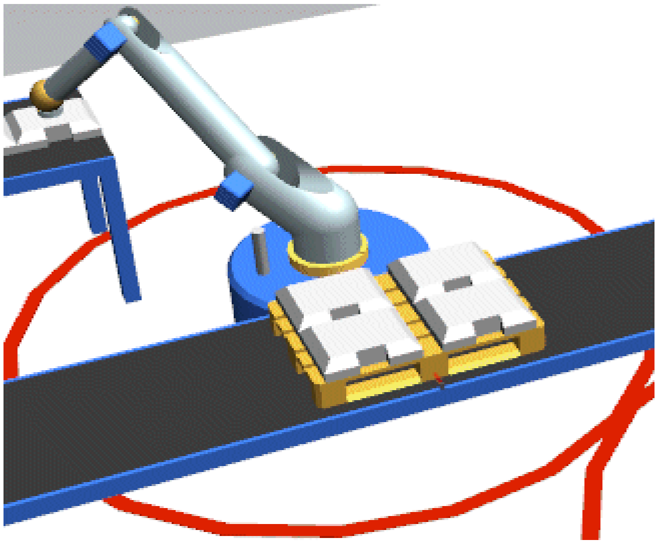


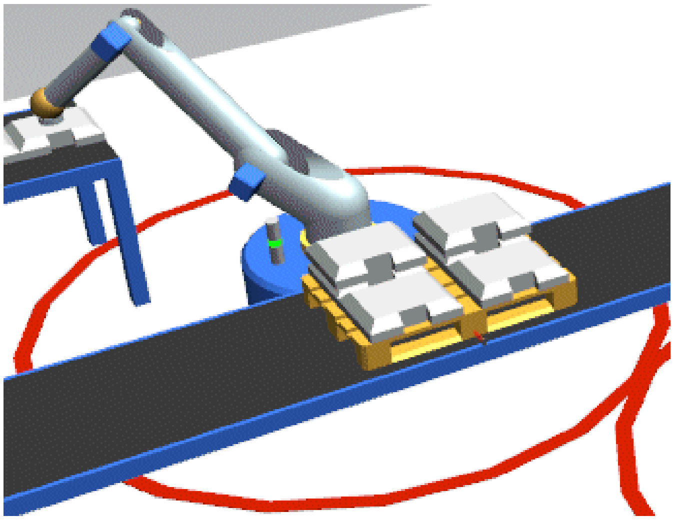


With the default setting **Fill Whole Layer** ☐ Fill Whole Layer deactivated *Plant Simulation* stacks the parts on the *Container*, before starting a new layer. This looks like this.



To load the *Container* layer by layer, activate **Fill Whole Layer** ☑ Fill Whole Layer . Once a layer is loaded all the way, *Plant Simulation* starts a new layer. This looks like this.





Back to **Model the Flow of Materials, Basics**

Working with Workpiece Carriers

Work with Workpiece Carriers

You can use the *Container* as a *workpiece carrier*.

To do so, select the check box in its dialog. Then *Plant Simulation* does not consider the **Workpiece Carrier** itself, but the **workpiece** loaded onto it, i.e., the *MU*. This affects, among others, the **Processing Time** and the **Set-up Time** of the part.

We demonstrate the *workpiece carrier* in connection with an *AssemblyStation* for which the main part is attached to a *workpiece carrier*.

In our example two *Sources* produce a *Container* each which we will use as *workpiece carriers*. In the **Entrance Control** of the *Sources* we produce an additional *Container* each, our *main part*, which is attached to our first *Container*, the *workpiece carrier*.

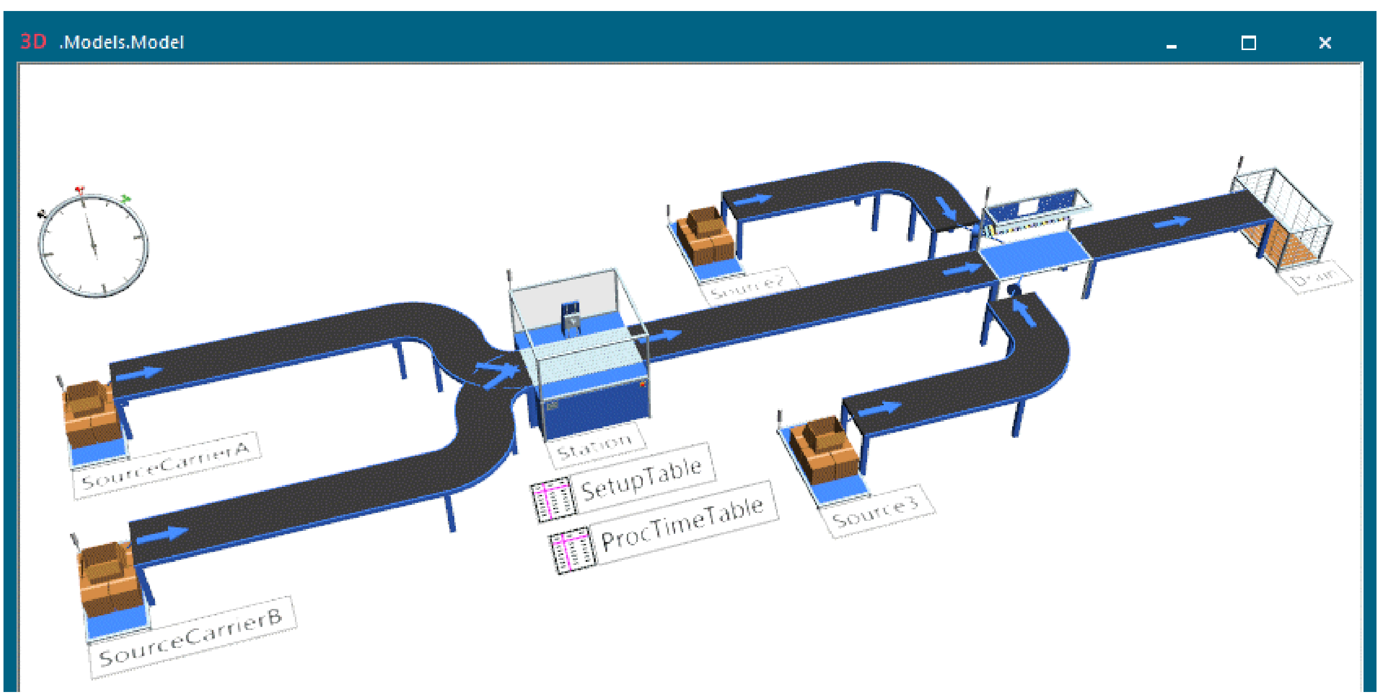
Two *Conveyors* transport the *main parts* attached to the *workpiece carriers* to a *Station*, which is set up according to the *SetUpTable*, and which processes the parts according to the *ProcessingTable*.

The *Containers* are then delivered across an additional *Conveyor* to the *AssemblyStations*. The *AssemblyStation* is fed with two different *add-on parts* by two attached *Sources*. The *AssemblyStation* attaches the *add-on parts* to the *main parts* on the *workpiece carrier*. The *AssemblyStation* then moves the fully assembled part on to the *Drain*.

We will demonstrate how to:

- **Configure Workpiece Carriers, Main Parts, and Add-On Parts**
- **Configure the Sources for Workpiece Carriers and Main Parts**
- **Configure the Processing Station, the Set-Up and Processing Table**
- **Configure the Sources for the Add-On Parts**
- **Configure the AssemblyStation Which Assembles the Parts**
- **Run the Simulation and View the Results**

The finished simulation model looks like this:



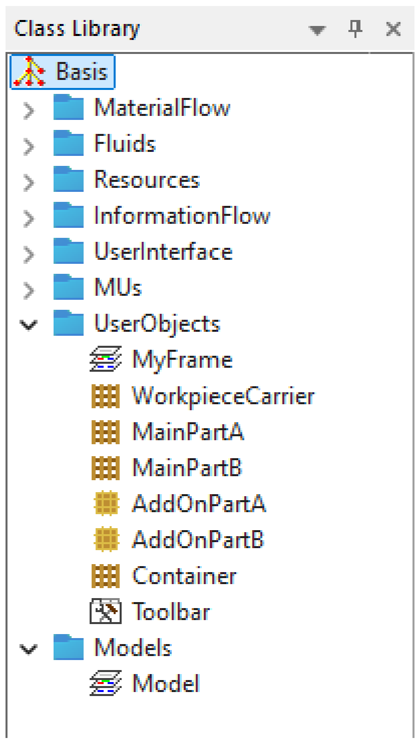
Back to Model the Flow of Materials, Basics

Configure Workpiece Carriers, Main Parts, and Add-On Parts

For our *workpiece carrier* and for our *main parts* we use the *Container*.

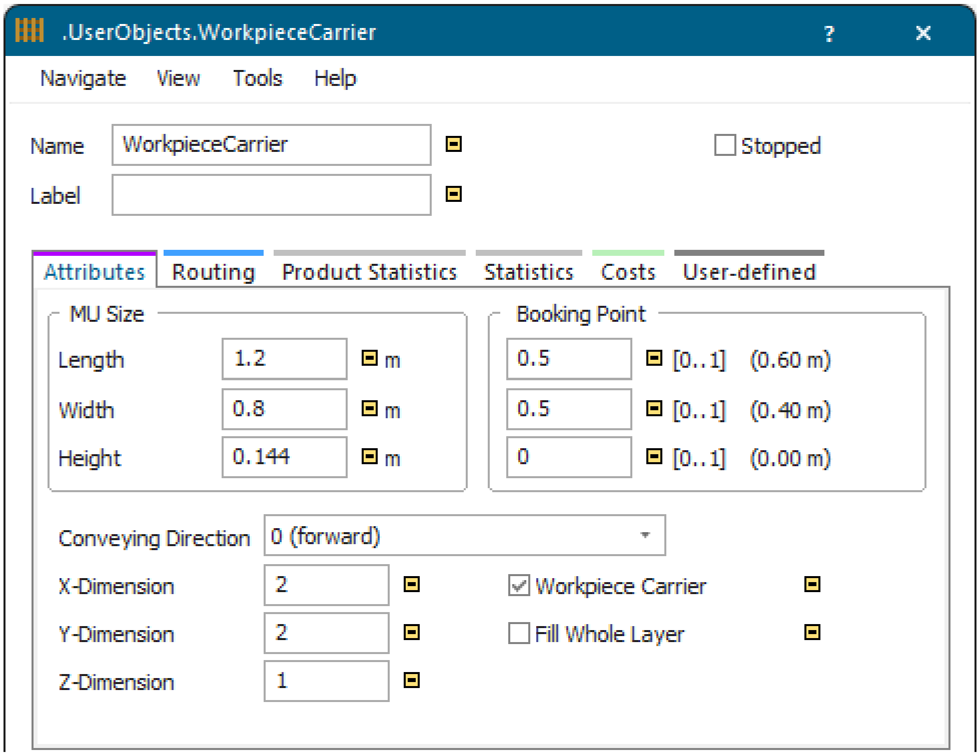
Proceed as follows:

- Open the folder **MUs** in the *Class Library*.
- Click the object *Container* with the right mouse button and select **Duplicate**.
- Rename the duplicated *Container* in the folder **UserObjects** to `WorkpieceCarrier`.
- Duplicate the *Container* twice and rename the copies to `MainPartA` and `MainPartB`.
- Duplicate the *Part* twice and rename the copies to `AddOnPart` and `AddOnPartB`.

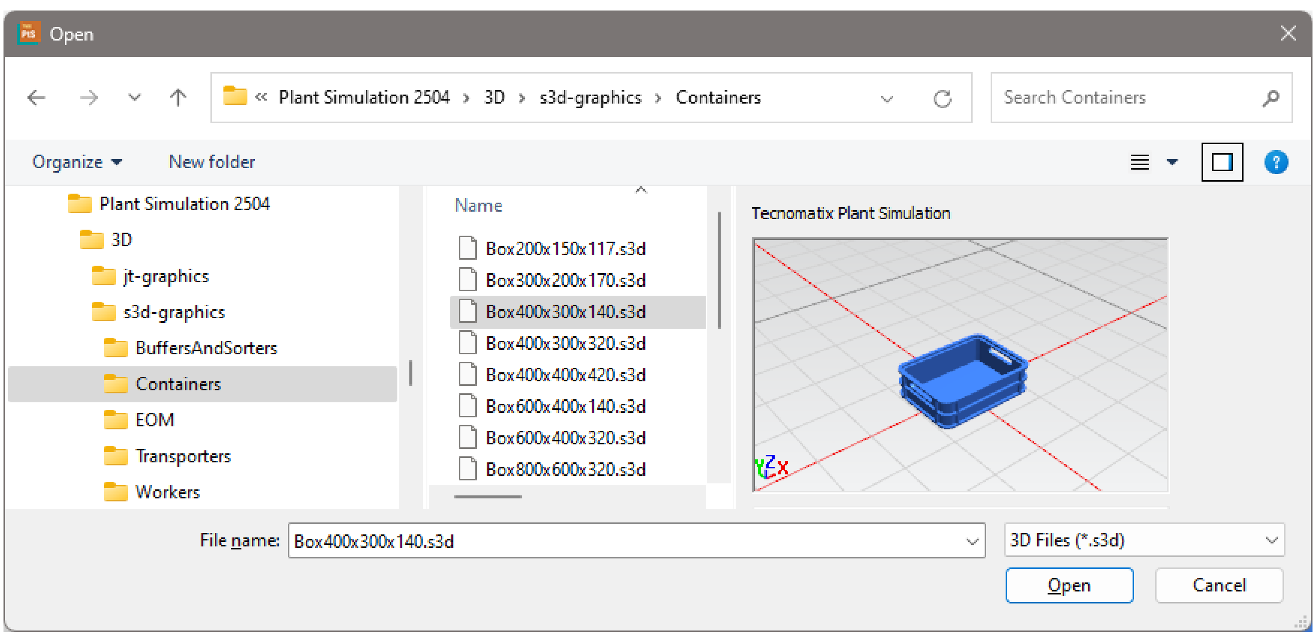


Configure the *WorkpieceCarrier*.

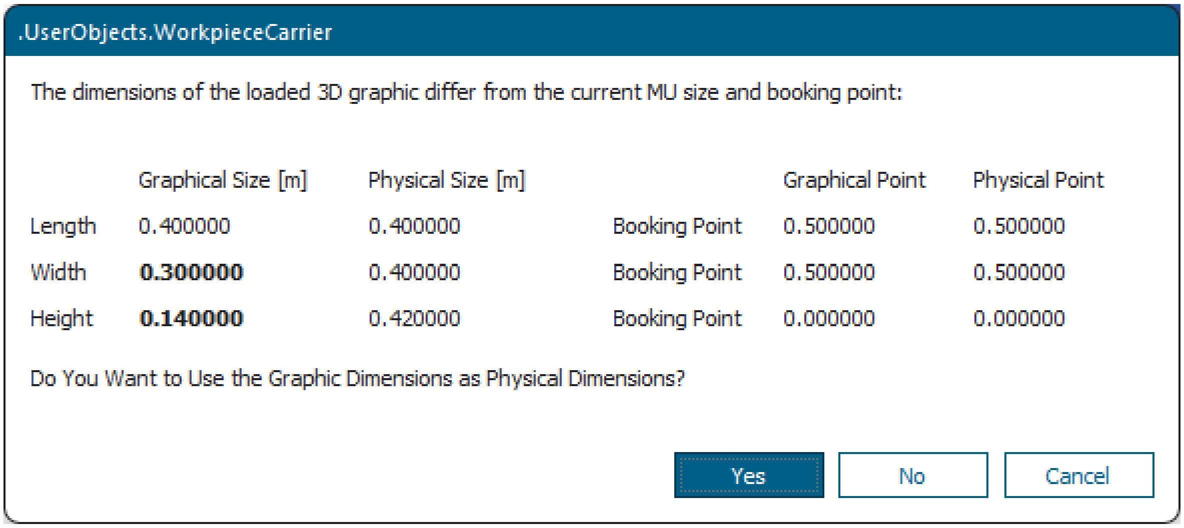
- We selected these settings for the **simulation object**:



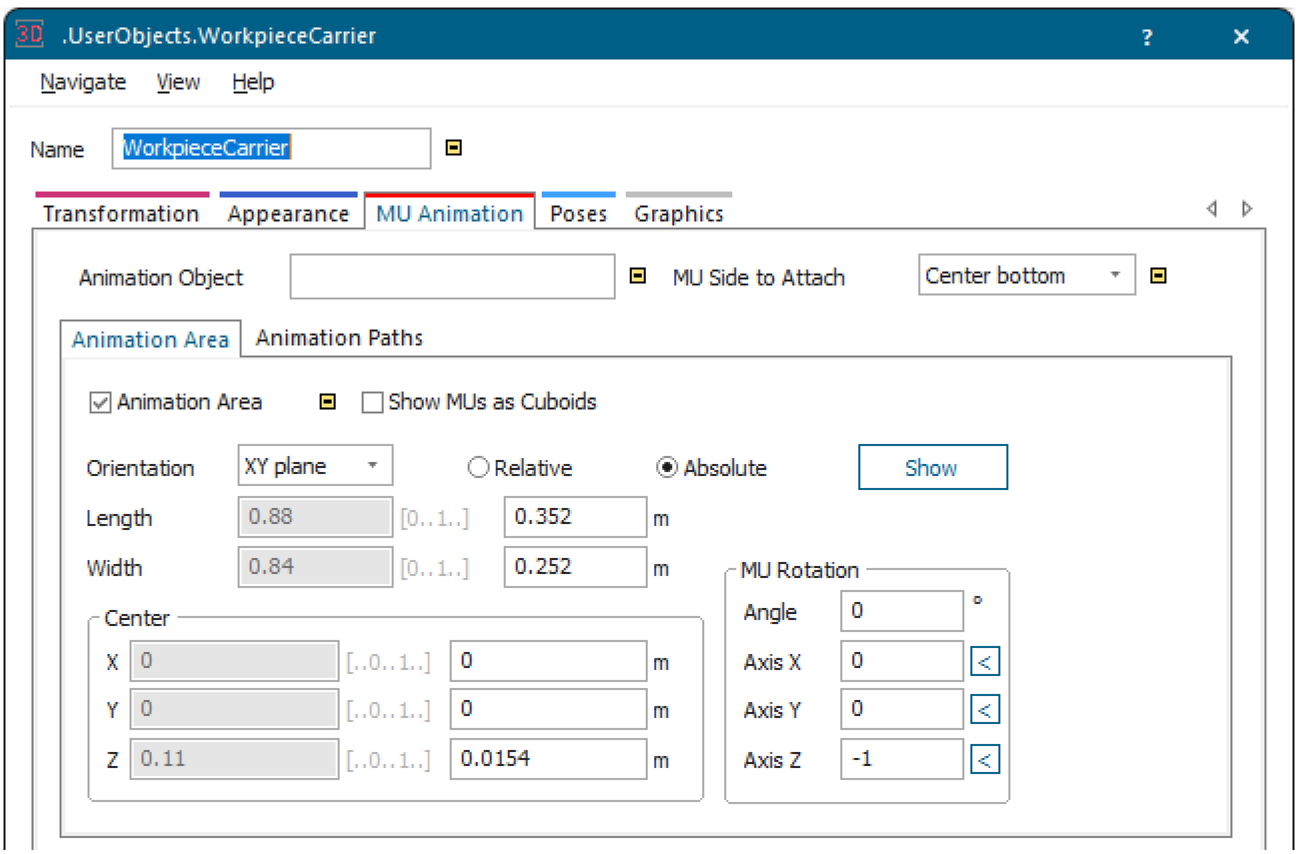
- Configure the **animation object** of the *WorkpieceCarrier*.
  - Click the *WorkpieceCarrier* with the right mouse button and select **Open in 3D**.
  - Click the right mouse button in the background of the 3D window and select **Exchange Graphics**. We selected this stacking box:



  - *Plant Simulation* then shows this message. As we do not want to use the suggested dimensions of the stacking box, we click **No**.

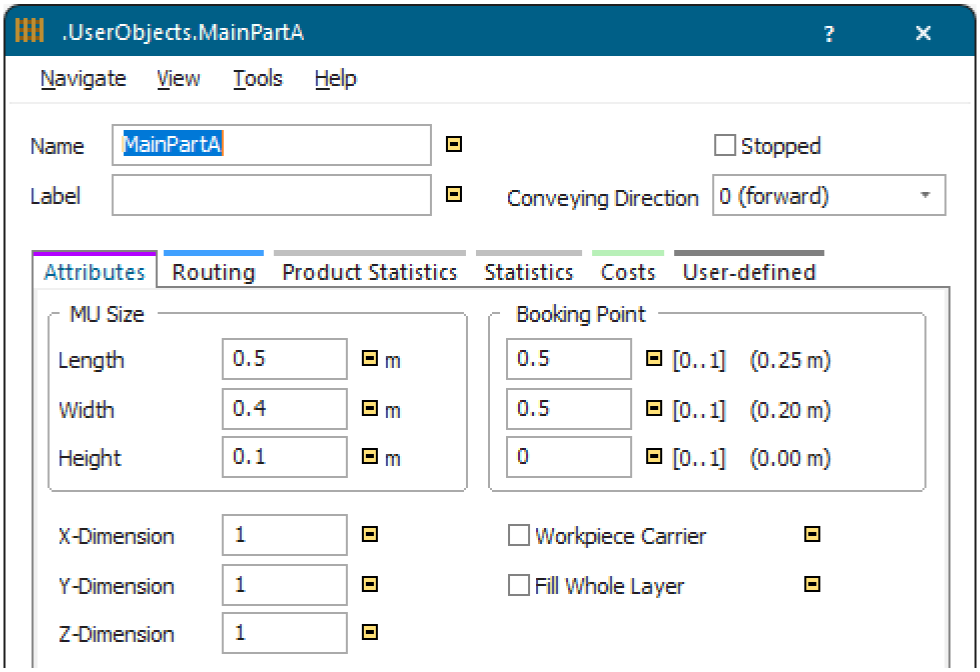


The **3D Properties** for the **MU Animation** of the *WorkpieceCarrier* look like this:

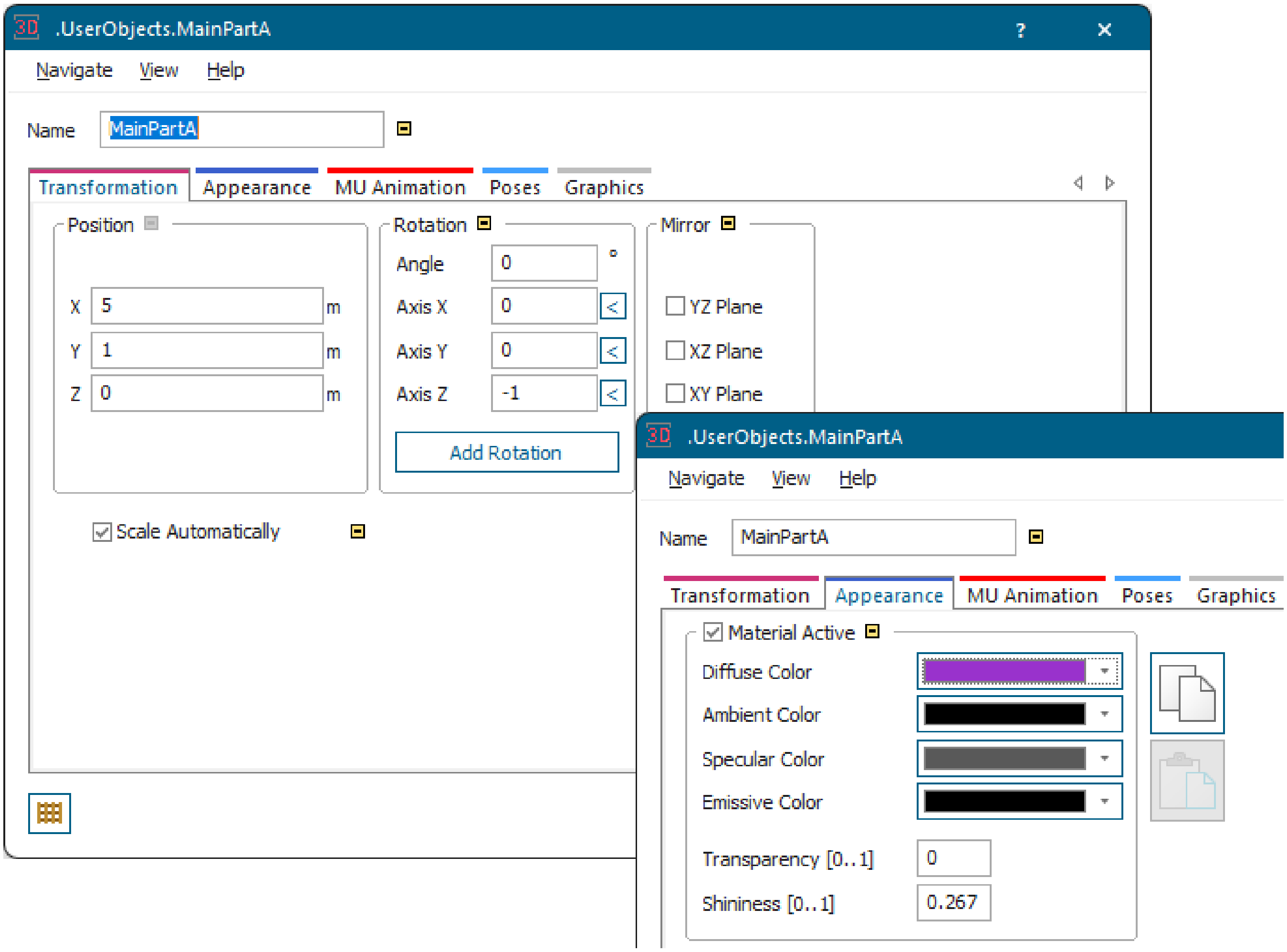


Configure the *MainParts*:

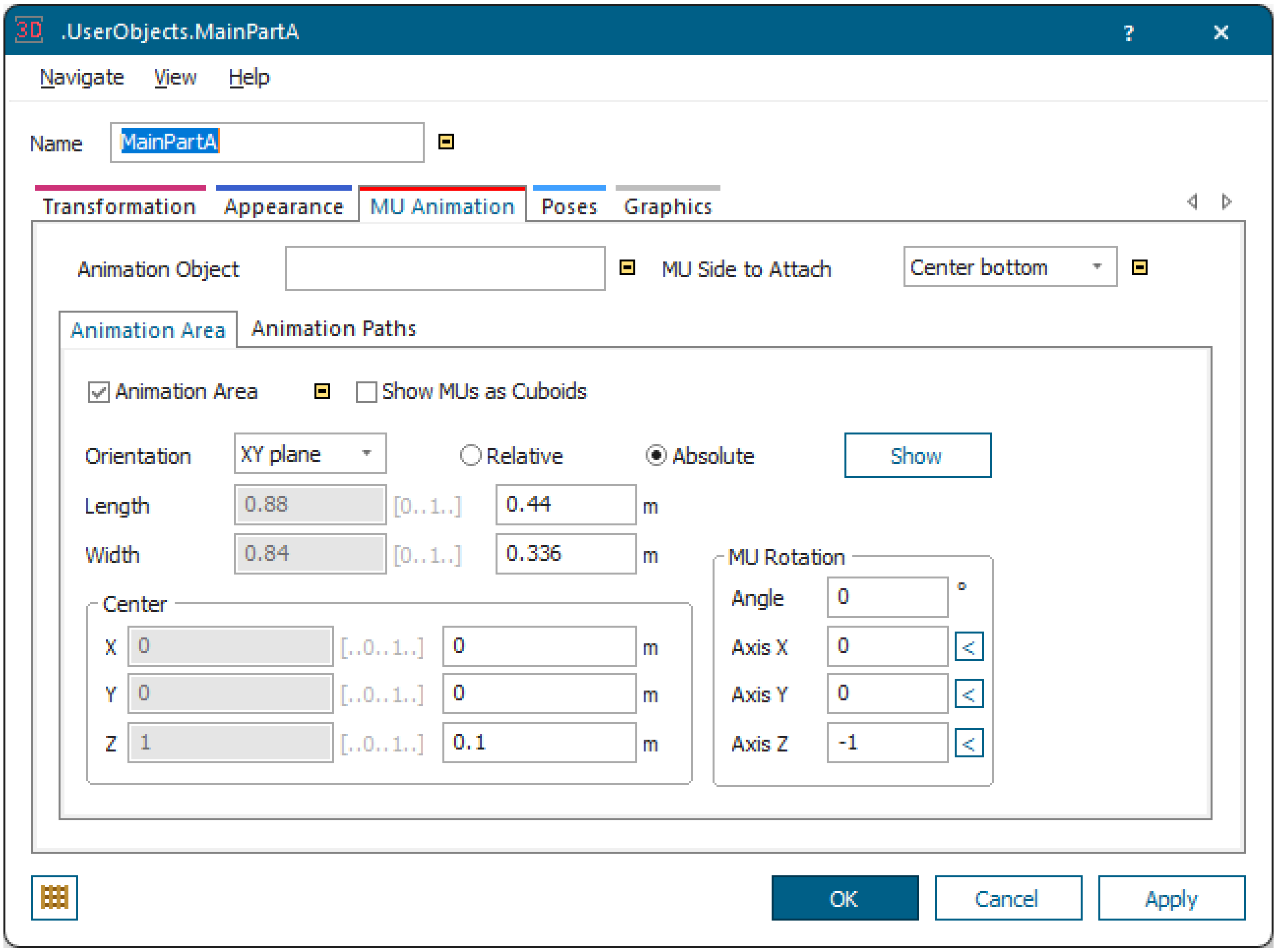
- As we want to attach the *MainParts* to our *WorkpieceCarriers*, we have to make them a little smaller than the blue stacking box. We copied the default graphic of the *Part* and pasted it into the main parts. The settings for the **simulation objects** of *MainPartA* and *MainPartB* now look like this:



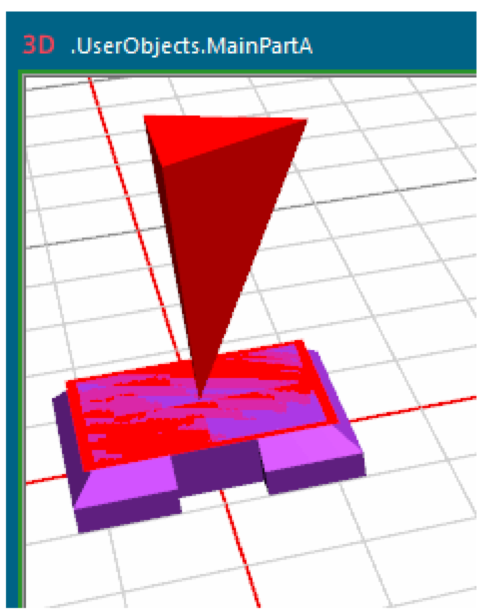
- For the **animation objects**of *MainPartA* and *MainPartB* we selected these settings for the **Transformation** and the **Appearance**. For *MainPartB* we just use another color.



For the **MU Animation** we use an **animation area** with these settings.

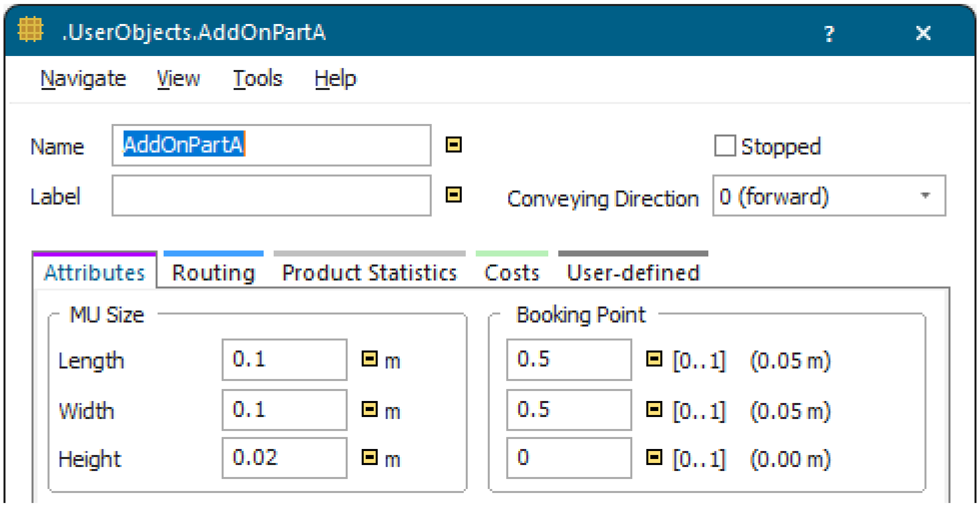


To make sure that the add-on parts are shown on the top side of the **animation area** instead of on its floor, we click **Show**. The setting **Center > Z > 1** ensures this.



Configure the **add-on parts**:

- For our *AddOnParts* that we want to attach to the *MainParts* we delete the default graphic and create a simple cylinder.
- We selected these settings for the **simulation object** and for the graphic of *AddOnPartA* and *AddOnPartB*. For *AddOnPartB* we just use another color.



Our add-on part A looks like this:

We did not change any other settings.

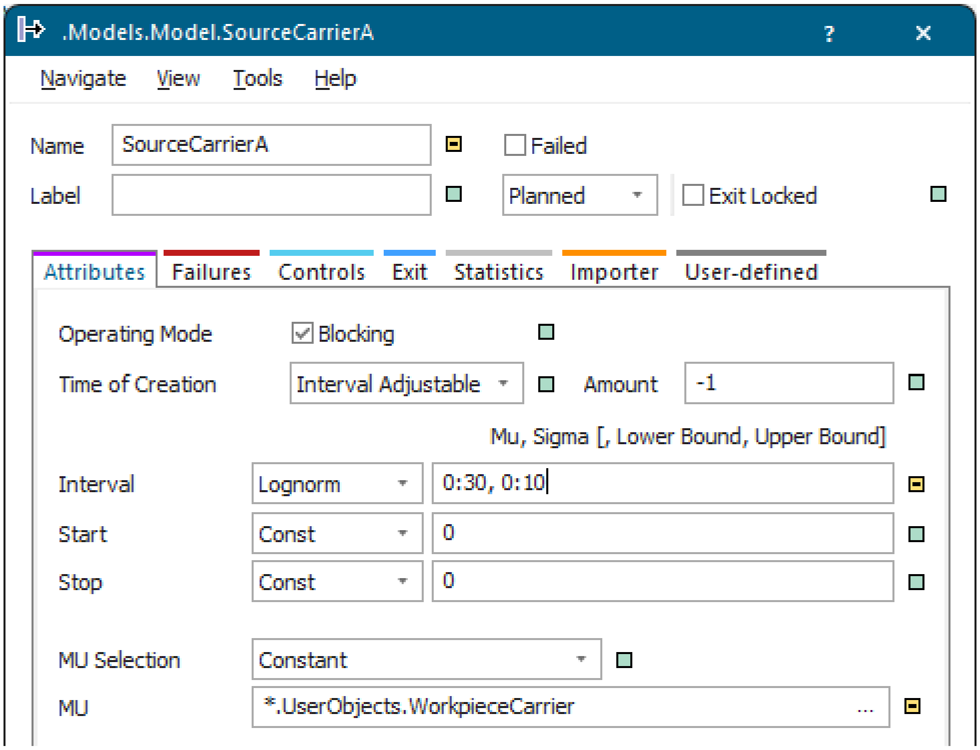
Back to Work with Workpiece Carriers

Configure the Sources for Workpiece Carriers and Main Parts

You can configure the *Sources* for creating *workpiece carriers* and main parts.

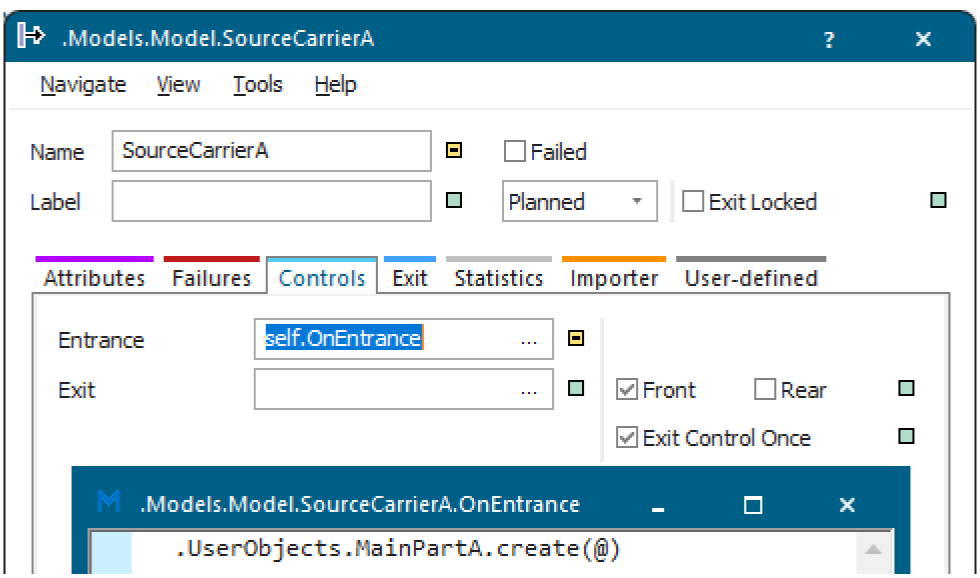
In this modeling step we configure the *Sources* which produce the *WorkpieceCarriers*:

- Our *SourceCarrierA* produces the *WorkpieceCarriers* for the *MainPartA* with these settings.



- In addition, we created an **Entrance Control** which creates a *MainPartA* on each *WorkpieceCarrier*. We typed in this instruction:

```
.UserObject.MainPartA.create(@)
```



- For the *SourceCarrierB* we use the same settings. We do, however, create a *MainPartB* in the **Entrance Control** with this instruction:

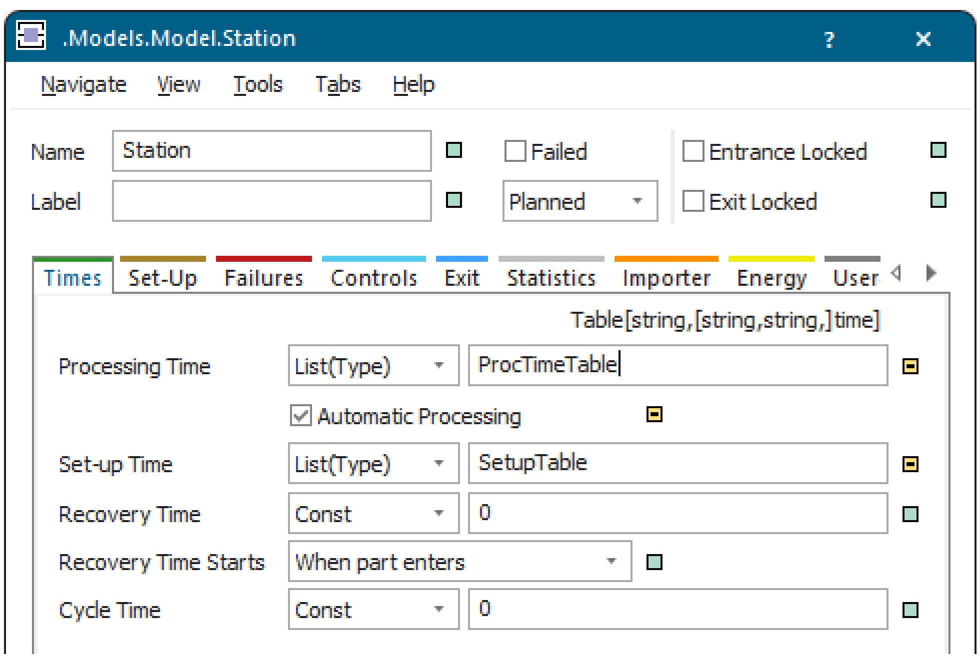
```
.UserObject.MainPartB.create(@)
```

Back to Work with Workpiece Carriers

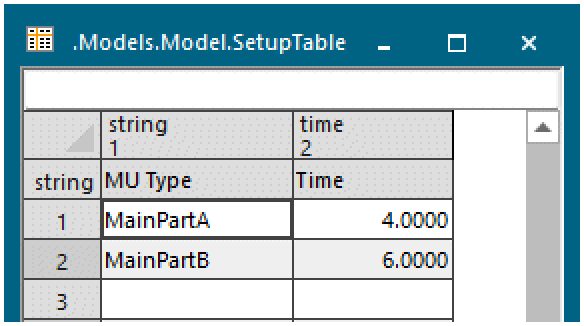
Configure the Processing Station, the Set-Up and Processing Table

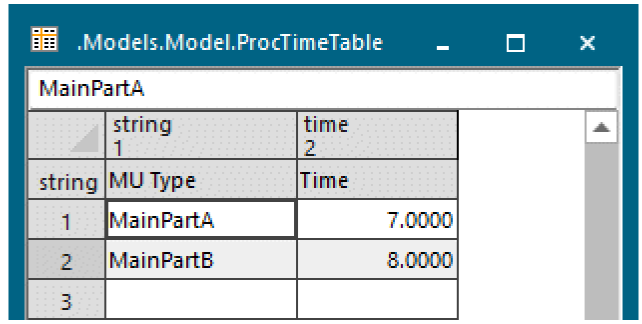
In this modeling step we configure our *ProcessingStation* which processes the *MainParts* on the *WorkpieceCarriers*.

- We set the **Set-up Time** for the *MainParts* on the *Station* in a *DataTable* of type **List(Type)**.
  - Insert a *DataTable* and rename it to `SetUpTable`.
  - Select **Set-up Time > List(Type)**, drag the *SetUpTable* to the text box next to it and drop it there.
- We also set the **Processing Time** of the *MainParts* on the *ProcessingStation* in a *DataTable* of type **List(Type)**.
  - Insert a *DataTable* and rename it to `ProcessingTable`.
  - Select **Processing Time > List(Type)**, drag the *ProcessingTable* to the text box next to it and drop it there.



- *Plant Simulation* formats the *DataTables* automatically with two columns, one for the **MU Type** of the part to be set-up for/processed, and one for the **Processing Time/Set-up Time**. We entered these **Processing** and **Set-up Times** for our *MainParts*:



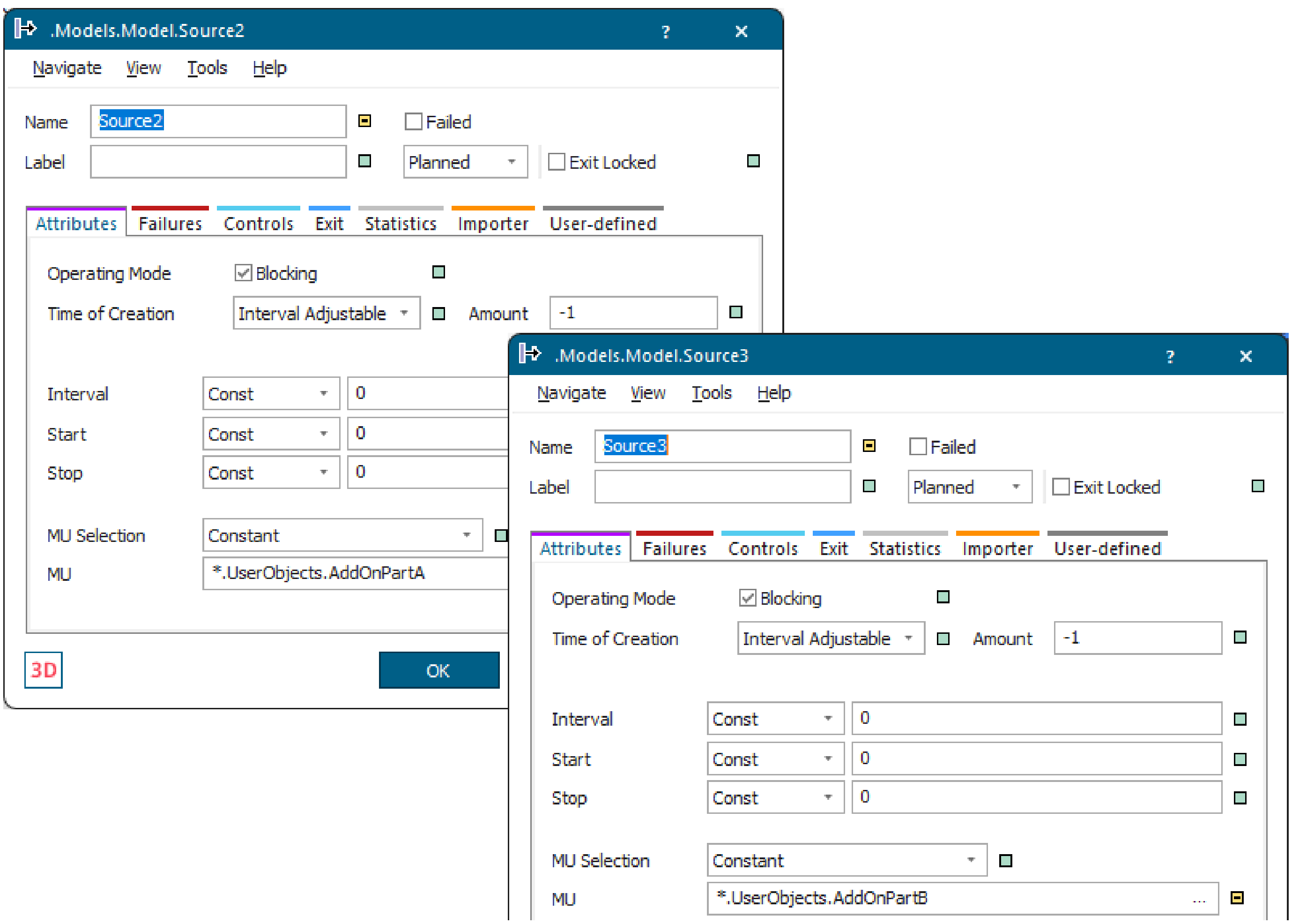


Back to [Work with Workpiece Carriers]

Configure the Sources for the Add-On Parts

In this modeling step we configure the *Sources* which feed the *AddOnParts* to the *AssemblyStation*, which then attaches these to the *MainParts*.

We selected our *AddOnPartA* and our *AddOnPartB* as the **MU** to be created.



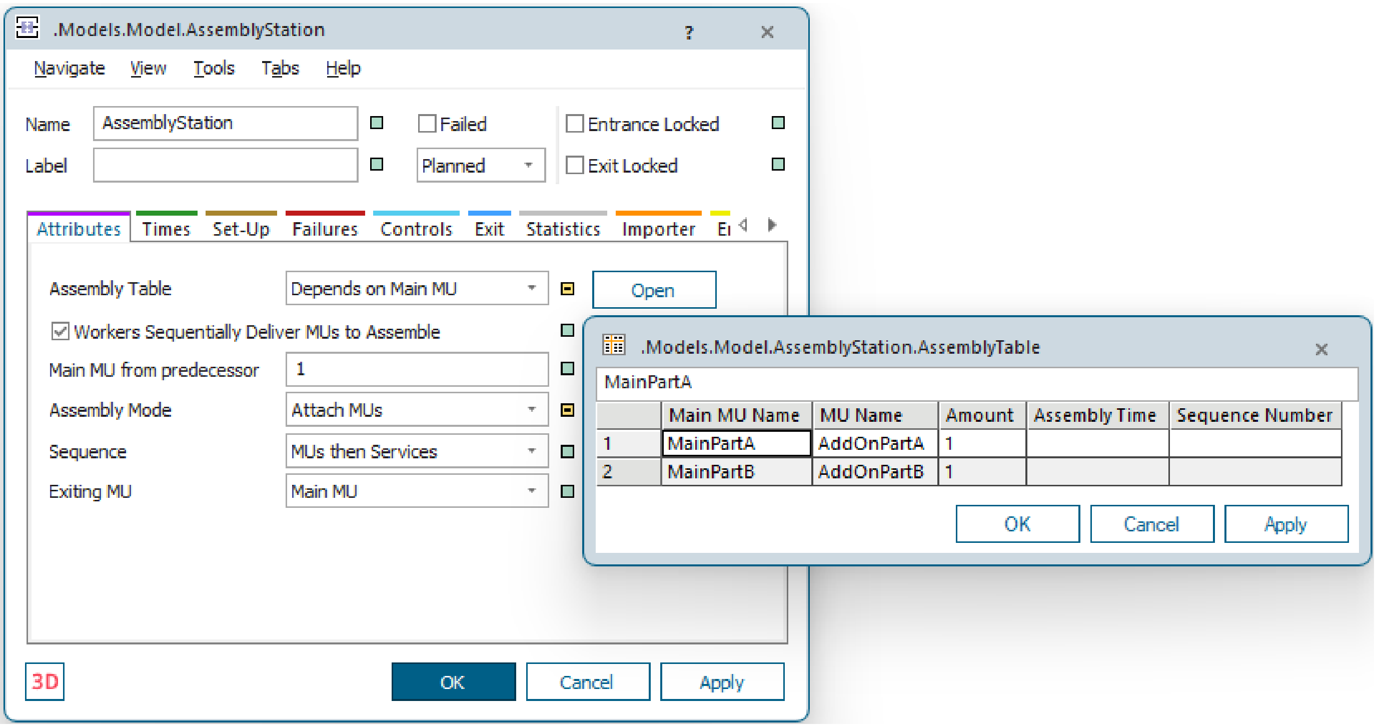
We did not change any other settings.

Back to Work with Workpiece Carriers

Configure the AssemblyStation Which Assembles the Parts

In our last modeling step we configure the *AssemblyStation* which attaches the *AddOnParts* to the *MainParts*.

To do so, we selected the settings below and entered into the *AssemblyTable* that we want to attach the respective *AddOnPart* to the associated *MainPart*.



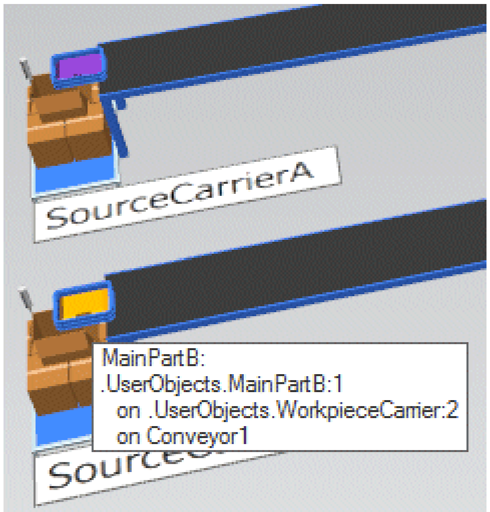
We did not change any other settings.

We finally insert a *Drain* and connect the individual material flow objects with *Conveyors* and with *Connectors*. For the *Drain* we entered a constant processing time of 1 minute.

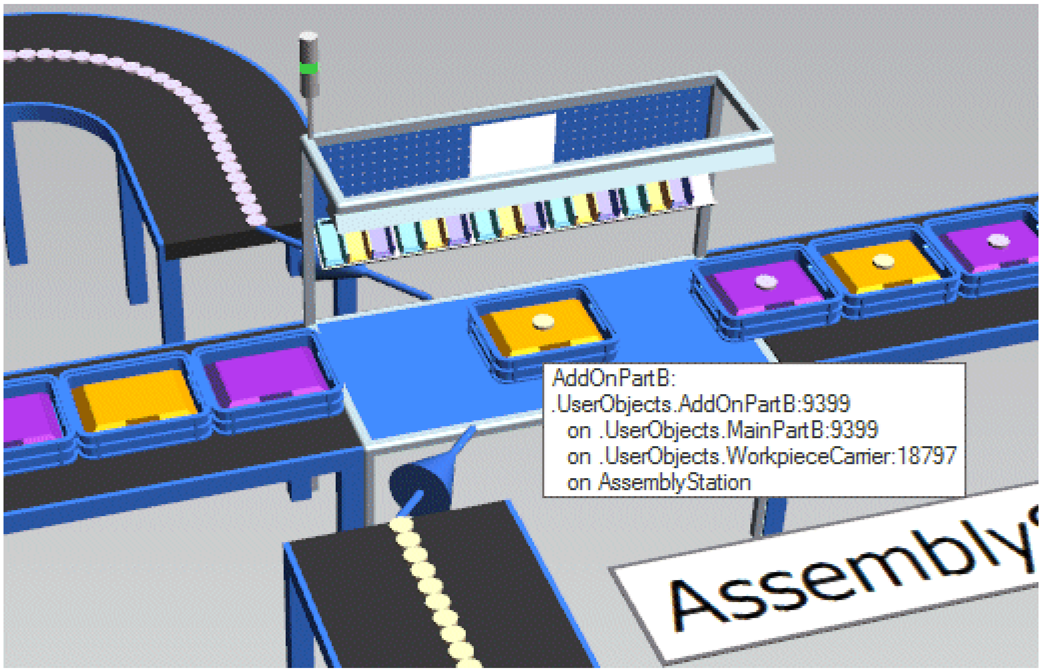
Back to Work with Workpiece Carriers

Run the Simulation and View the Results

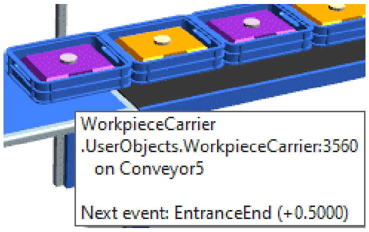
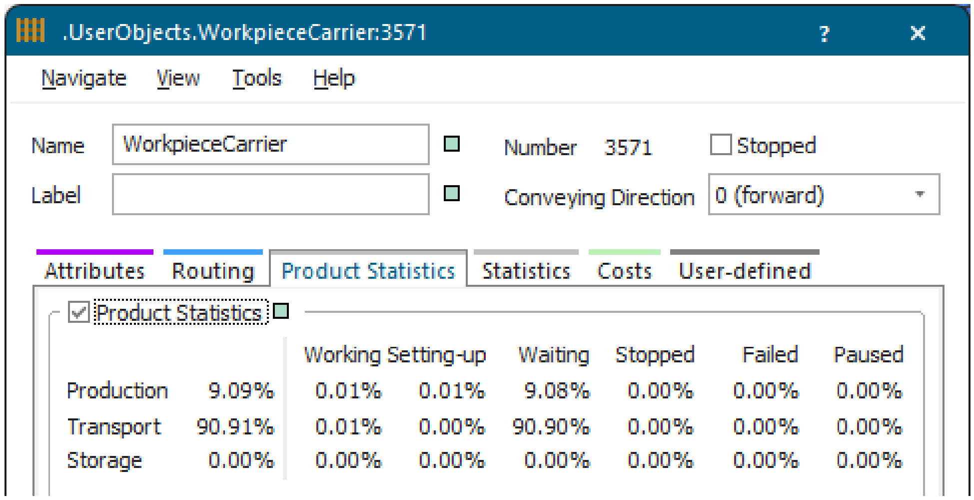
Run the simulation and watch that the *Sources* first create our *WorkpieceCarriers*, i.e., the blue stacking boxes, and then the respective *MainParts*, i.e., the violet or orange plates.

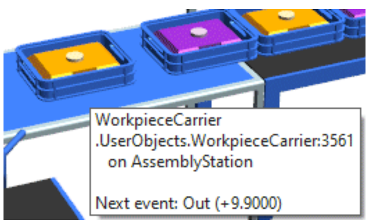
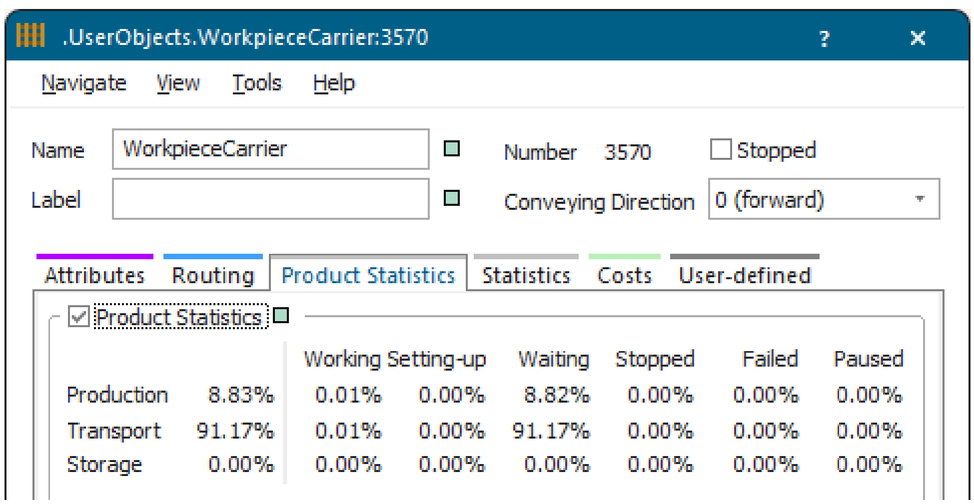


After the *MainParts* were processed on the *ProcessingStation*, the *AssemblyStation* attaches the *AddOnParts* to the *MainParts* which are located on the *WorkpieceCarriers*.



Product Statistics of our *WorkpieceCarrier* for the attached *MainPartA* or *MainPartB* looks like this after it has left the *AssemblyStation*:

<table>
<tr><th>WorkpieceCarrier MainPartA</th><th>Product Statistics WorkpieceCarrier MainPartA</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

<table>
<tr><th>WorkpieceCarrier MainPartB</th><th>Product Statistics WorkpieceCarrier MainPartB</th></tr>
<tr><td>
</td><td>
</td></tr>
</table>

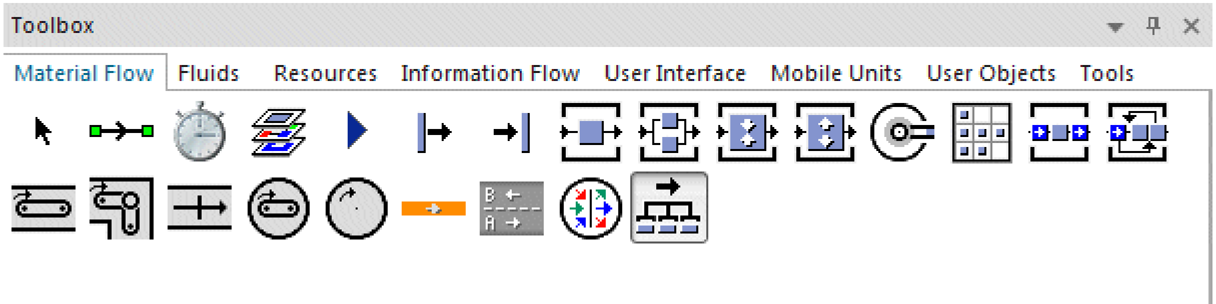
Back to Work with Workpiece Carriers

Balancing a Production Line

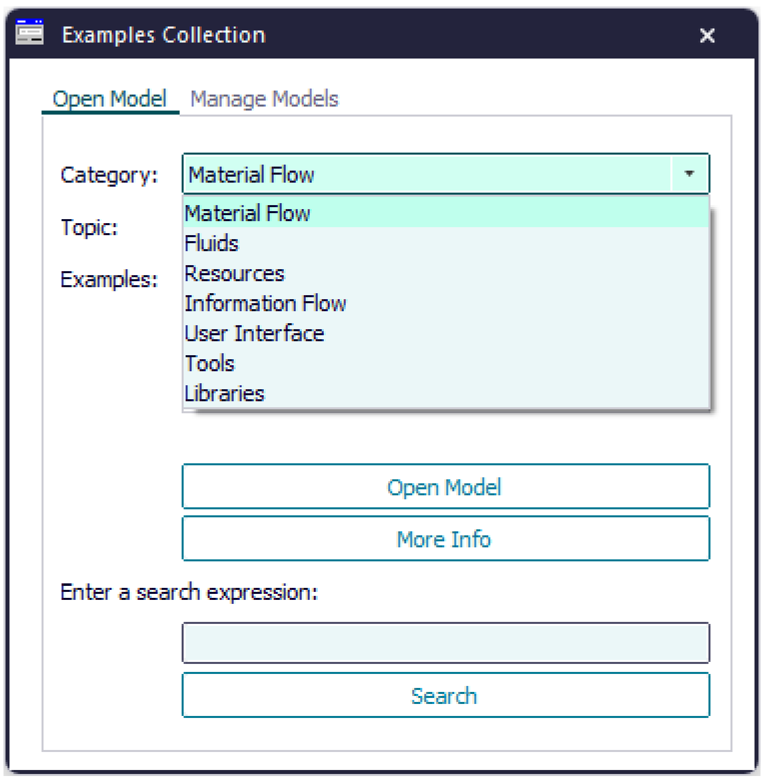
Balance a Production Line

The object *Cycle* synchronizes the transfer of *parts* from station to station within a production line. You can use it to only move a part on to the next station within a balanced line, when all stations have finished processing their parts and when none of the stations is failed, paused, or unplanned. In addition, the successor of the balanced line has to be ready to receive the part.

You can insert the object **Cycle** into your simulation model from the folder **MaterialFlow** in the *Class Library* or from the toolbar **MaterialFlow** in the *Toolbox*.



Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.



We will demonstrate how to:

- Open the Production Line Model Created in 2D
- Convert the Production Line Model to a 3D Model

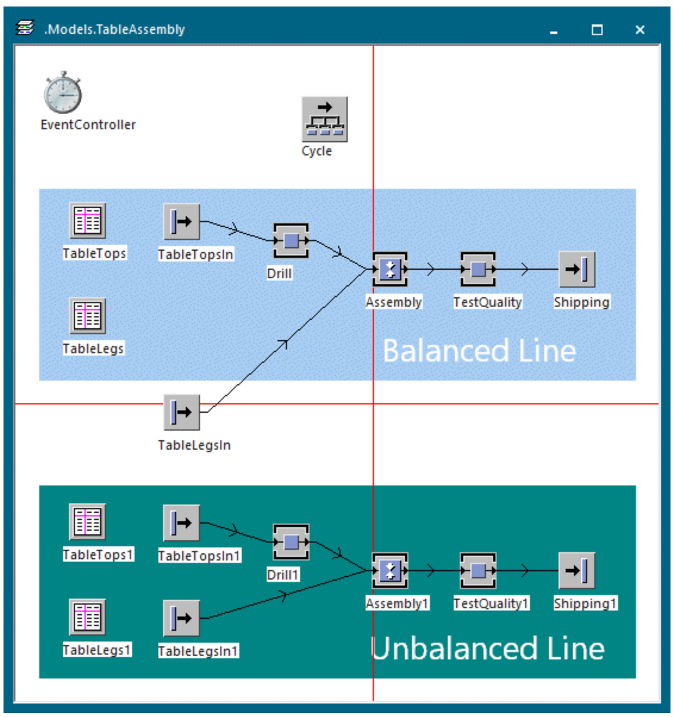
Back to Model the Flow of Materials, Basics

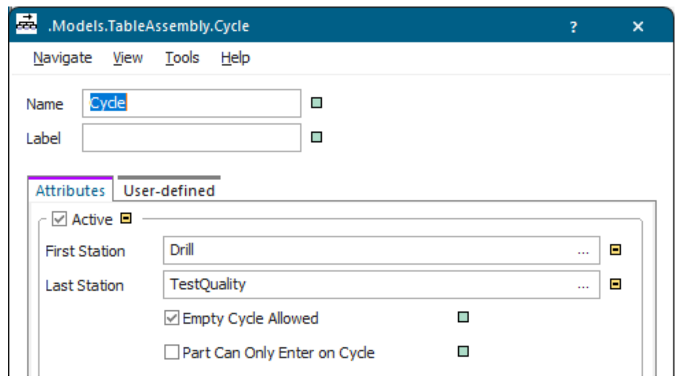
Open the Production Line Model Created in 2D

We open the simulation model in 3D, which we created in a previous version in 2D only mode.

We entered the name of the **first station** and of the **last station** into the text boxes to define the balanced line. All stations between the first and the last station, which are connected with *Connectors*, form the balanced line. Each station has to have a predecessor and a successor and can only have a single predecessor and a single successor.

> **Note**
>
> You can only balance production lines that consist of objects of type *Station* and *AssemblyStation*. When an *AssemblyStation* is part of the balanced line, the *Cycle* only continues balancing, when the assembly process has been finished.





If you did not define any stations for the balanced line yet, you can drag an object of type *Station* or *AssemblyStation* onto the icon of the object *Cycle* and drop it there, *Plant Simulation* enters it as the first station of the balanced line.

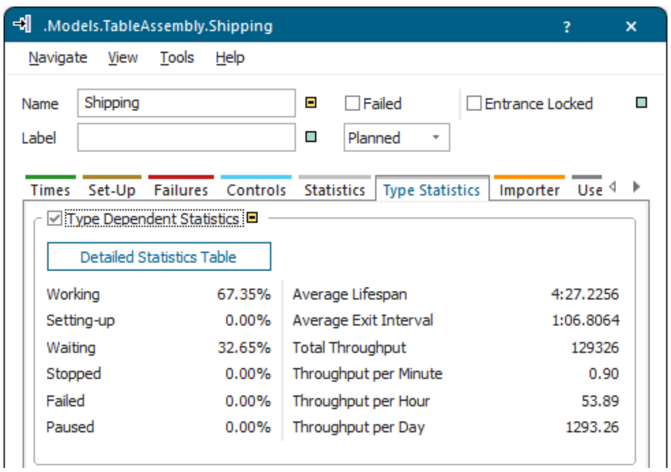
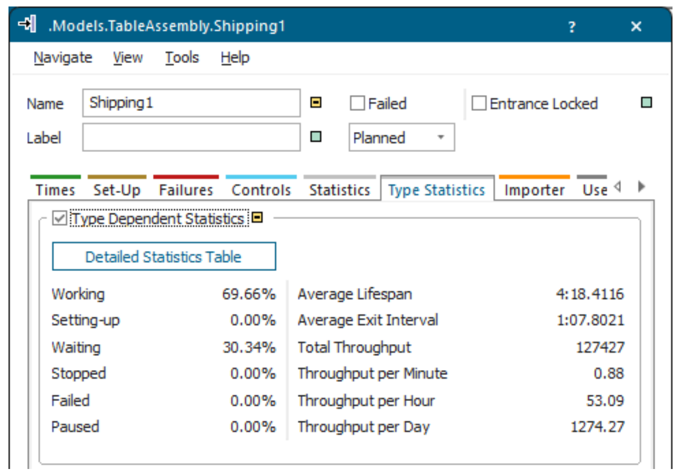
We did this with the station called *Drill*, which is a modified *Station*.

Then drag another object onto it and drop it there. *Plant Simulation* enters it as the last station.

We did this with the *AssemblyStation*. We then changed our mind and decided to make the station called *TestQuality* the last station. To do so, we held down **Shift**, dragged it over the *Cycle*, and dropped it there.

> **Note**
>
> As each station within the balanced line can only have a single predecessor and a single successor, the *Source* named *TableLegsIn*, that feeds the table legs to the *Assembly* station in our example above, cannot be part of the balanced line.

To detect the difference between a balanced line and an unbalanced line, we copied the balanced line and pasted it below it. We then started the simulation and checked the tab **Type Statistics** of the *Drain* of the balanced and of the unbalanced line.

Balanced line	Unbalanced line
	

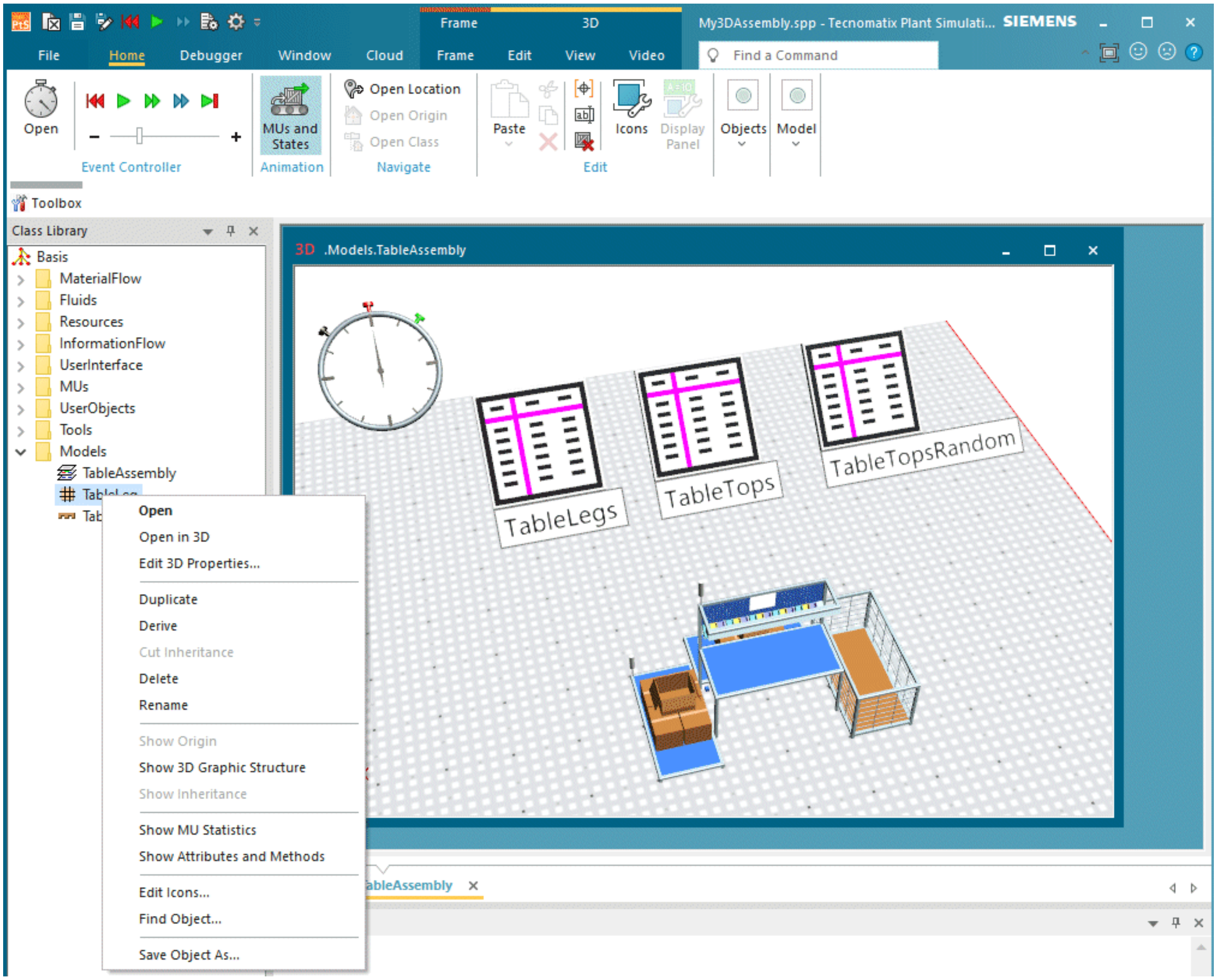
Back to Balance a Production Line

Convert the Production Line Model to a 3D Model

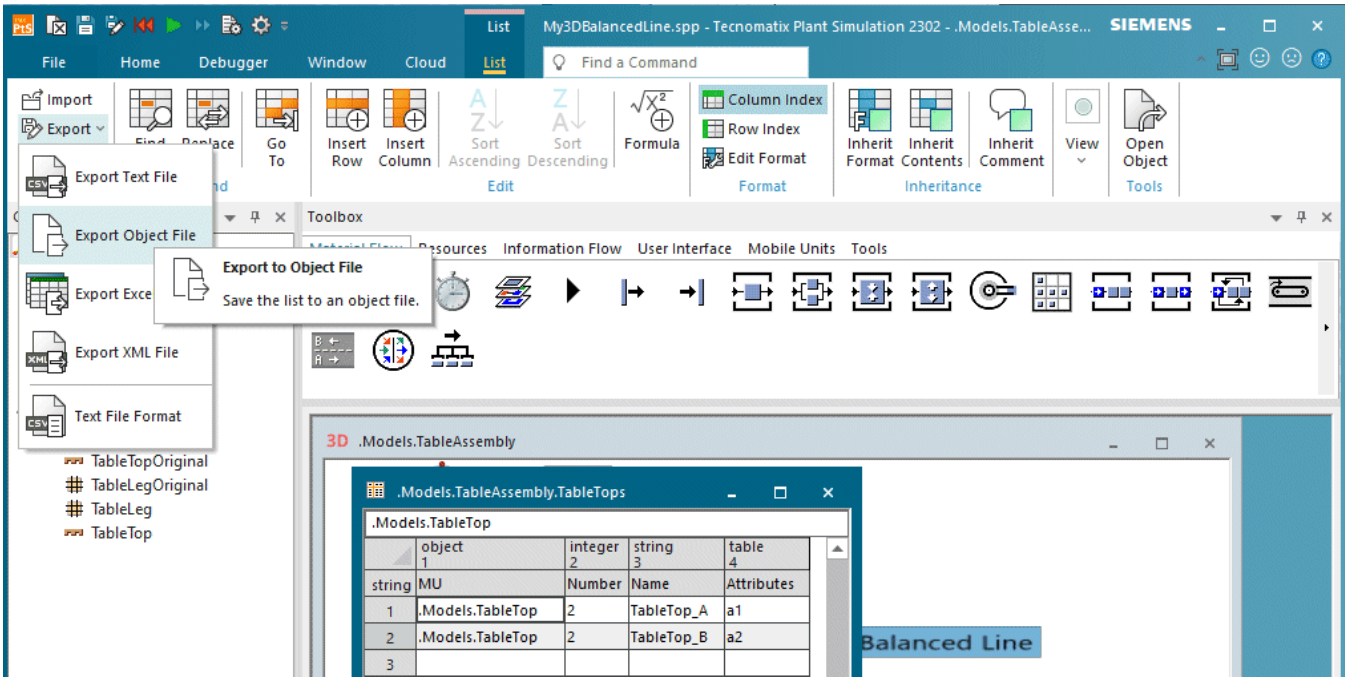
Here we show how to convert the model, that we originally created in 2D, to a 3D model.

Here we show how to convert the model, that we originally created in 2D, to a 3D model. As we already converted the *table assembly*, we can use the adjustments that we made in that model. Compare **Assemble Parts with the AssemblyStation** and **Adjust the Graphics of the Assembly**. To do so, we opened the model **My3DAssembly.spp**.

- We save the *MUs* named *TableLeg* and *TableTop* as an object file each with the name `TableLeg.psobj` and `TableTop.psobj`, compare **Save Object As**. These files contain the simulation properties and the 3D graphic settings that we defined.



- We open the *DataTables* named *TableLegs* and *TableTops*, select **Export Object File**, and export them as an object file each with the names `TableLegTable.psobj` and `TableTopTable.psobj`. These files contain the attributes that set with which color the table legs and table tops are shown.



Then we open our 2D model **MyBalancedLine.spp** that we created in a previous version of *Plant Simulation*.

- We save it with a new name. We typed in `My3DBalancedLine.spp`.
- We select the **Visualization > 3D only** in the **Model Settings**.

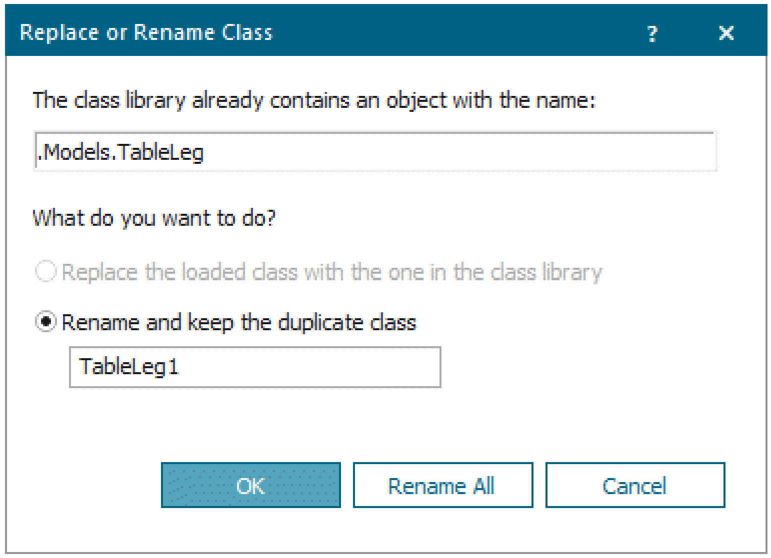
> **Note**
>
> If some objects, for example the *DataTable*, are not displayed in the 3D model, select this object in the *Class Library*, press **F8**, and double-click the attribute **CreateIn3D** so that it shows **true**.

- To use the graphics of the current version of *Plant Simulation*, we hold down **Shift**, click **Basis** with the right mouse button in the *Class Library*, and select **Revert all Objects to Standard Graphics**.

Plant Simulation then runs the simulation with the default settings of the parts. As this is not what we want, we will:

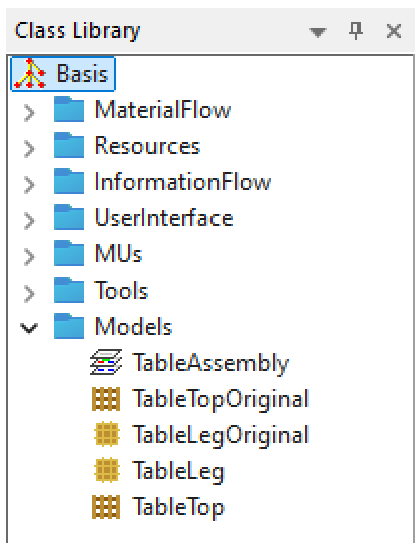
- **Import the simulation settings and the graphics of the table legs and the table tops**

  Click the folder **Models** in the *Class Library* with the right mouse button and select **Save/Load > Load Object**. Select the file **TableLeg.psobj** which we exported above. Repeat this for the **TableTop.psobj**. Click **OK** in the dialog that opens.



Rename the original classes, for example to `TableLegOriginal` and to `TableTopOriginal`, and the imported classes to `TableLeg` and to `TableTop`. Once everything works as expected, we can delete the original classes.

The *Class Library* then looks like this.

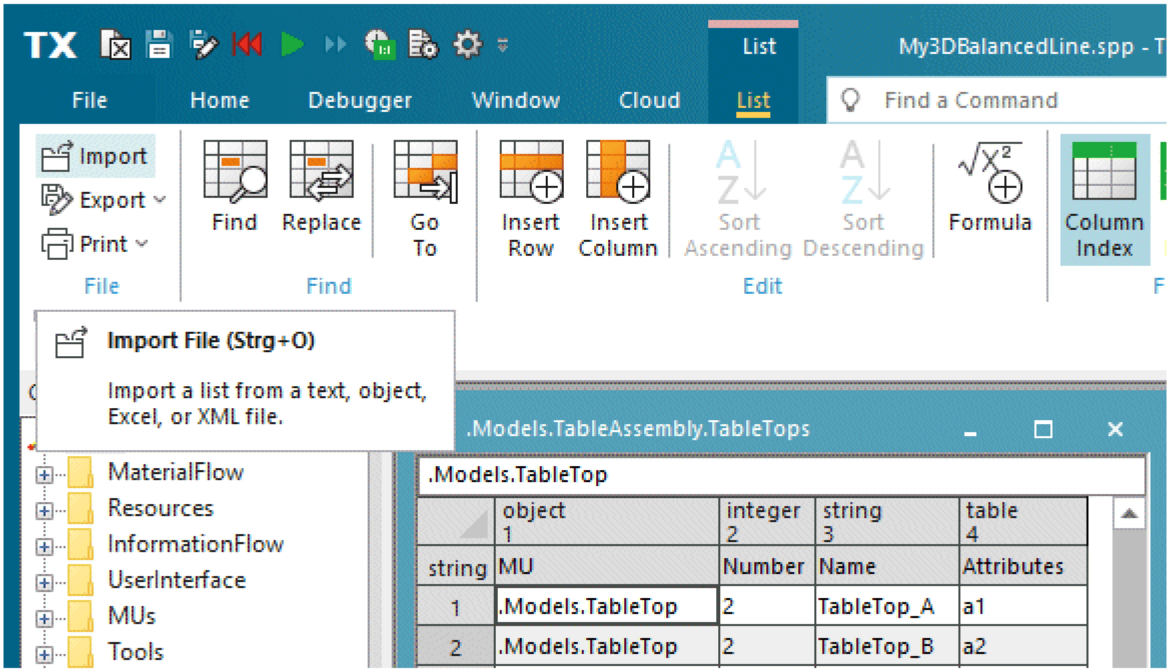


When we run the simulation, we see that the table tops and the table legs are produced and assembled as expected. The colors of the table tops and the table legs are wrong though, as these are set in the production tables.

- **Import the contents of the production tables of the table tops and the table legs**

  Open the production tables *TableTops*, *TableLegs*, *TableTops1*, and *TableLegs1* in the *Frame TableAssembly*.

  Click **Import File** and select the respective file **TableLegTable.psobj** or **TableTopTable.psobj**, which we exported above.



Now the table tops and the table legs show the correct color. We finally adjust the positions of the objects and insert a *Comment* each showing `Balanced Line` and `Non-Balanced Line`.